Graphical Abstract

When does fusing hand-crafted knowledge with learned representations pay? A controlled, cost-normalized benchmark and a measurable rule for stacking, substitution and interference

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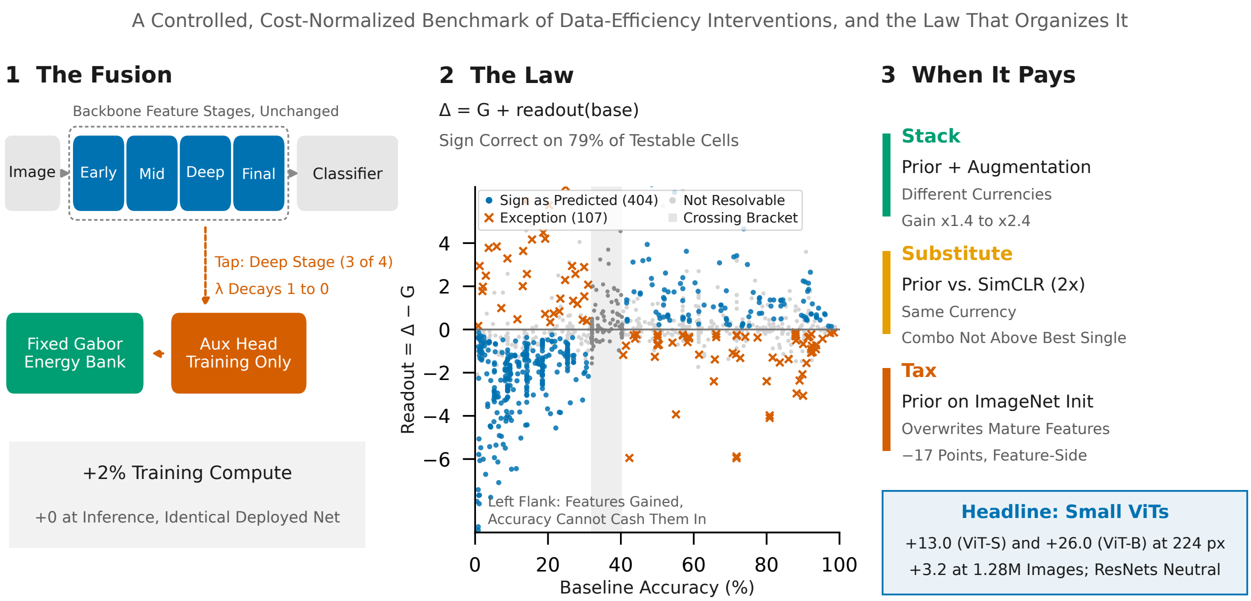

Highlights

When does fusing hand-crafted knowledge with learned representations pay? A controlled, cost-normalized benchmark and a measurable rule for stacking, substitution and interference

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- A 2,978-configuration benchmark of data-efficiency interventions
- Splitting one instrument's bands pays; pairing two satellites does not
- A measurable rule predicts stacking, substitution or interference
- Free prior beats 2×-compute SSL on small ViTs; at 5× the ordering flips with data
- Attention's feature deficit persists at 1.28M images and grows with scale

When does fusing hand-crafted knowledge with learned representations pay? A controlled, cost-normalized benchmark and a measurable rule for stacking, substitution and interference

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ABSTRACT

Fusing prior knowledge with data-driven learning is attractive where data is scarce, yet no controlled account says when such fusion helps, is redundant, or harms. We present a benchmark that measures one fixed, hand-crafted source of knowledge, a pinned bank of Gabor targets injected only during training at ∼2% compute overhead, against the data-driven alternatives (SimCLR, SimSiam and DINO pre-training, ImageNet transfer, heavy augmentation, learned teachers) under one frozen recipe with fixed subsets: 13 datasets, 9 backbones, 500 to 1.28M images, 32 to 224 px, and 5.7M to 86M parameters (2,978 configurations over 9,079 runs). Every pairwise combination is measured, and all follow one rule at training time (decision-level fusion behaves differently): different-*currency* sources stack (prior with augmentation), same-currency sources substitute (prior against effective self-supervised pre-training), and fusing into an already-informed initialization interferes in proportion to what it carries (ImageNet transfer, up to −17 points). Because each source's currency is measurable on frozen features beforehand, the outcome is predictable; we state this as a procedure. Two sensor tests bound it: splitting one instrument's bands pays at every data scale, whereas pairing two satellites (Sentinel-1 radar with Sentinel-2 optical) never does; the two settings differ in both source asymmetry and modality, and our design cannot say which is operative. Underlying it is an empirical decomposition, $\Delta = G + \text{readout(base)}$, whose sign holds on 79% of testable cells and which, registered in advance, called an unseen backbone family's feature gain from baseline heights alone. Code and data: amughrabi.github.io/MomentAux.

1. Introduction

A bank of Gabor filters, fixed before training and free to compute, is worth +26 accuracy points to a ViT-B/16 at 224 px. Injected exactly the same way into a network that has already had contrastive pre-training, it is worth nothing. Injected into one initialized from ImageNet, it costs 17 points. One intervention, one frozen recipe, one code path: three outcomes spanning forty accuracy points. The useful question is not which number is representative, because none of them is. It is what distinguishes the three cases, and whether that can be determined *before* the combination is trained.

This is the general situation, not a curiosity of one prior. Every practitioner who has trained a network on a small dataset faces the same menu of remedies: pre-train on ImageNet, run a self-supervised stage, augment more aggressively, or inject prior knowledge into the model. Each fuses an external source of information, weights learned elsewhere, invariances distilled from augmented views, hand-crafted structure, with whatever the labeled data can teach on its own. What the field lacks is not remedies but a controlled account of how these sources *combine*: when does adding a second source help, when is it redundant with what the first already supplies, and when does the fusion actively destroy value? These three outcomes, we will call them *stack*, *substitute*, and *tax*, are usually discovered anecdotally, one paper and one setting at a time, under training recipes that differ in a dozen uncontrolled ways.

Our answer is that sources trade in identifiable *currencies*, and that what a source is worth depends on which currency is scarce rather than on how modern the method is. Two sources add only when their currencies differ. This is not a metaphor we impose after the fact: each source's currency is measurable, on frozen features, from each source *alone*, before any combination is trained. That is what makes the three outcomes above predictable instead of anecdotal, and

Project page: https://amughrabi.github.io/MomentAux. Code, configurations, fixed subset indices, per-run records and all result tables: https://github.com/GCVCG/MomentAux

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it is why the same free prior can be the most valuable intervention we measure in one setting and the most damaging in another.

This paper treats the question as a measurement problem. We build a single instrument, one frozen training recipe, fixed data subsets shared by every run, and a numerically pinned bank of Gabor and moment spectral filters, and use it to measure one deliberately austere source of prior knowledge against the strongest data-driven alternatives at matched or declared compute. The prior, which we call *MomentAux*, could hardly be cheaper: a fixed bank of oriented-energy targets, injected only during training through a small auxiliary regression head whose loss weight decays to exactly zero, adding ∼2% training compute and *nothing* at inference. Against it we field SimCLR, SimSiam, and DINO initializations (each ∼2× compute), ImageNet transfer, DeiT-strength augmentation, learned FitNets-style teachers, and targets built from histograms of oriented gradients (HOG). The resulting grid spans 13 datasets across six visual domains, nine backbone families from ResNet-18 to ViT-B/16, data scales from 500 to 1.28M images, resolutions from 32 to 224 px, and 2,978 configurations trained over 9,079 runs, each with a retained run record.

A grid of this size threatens to be a table dump. It is rescued by an empirical law that we did not assume but found, and then tested the way laws should be tested, by registering its predictions in advance and trying to break them. Writing Δ for an intervention's end-to-end accuracy gain and G for the gain a linear evaluation reads from its frozen features under identical probing, we find across the grid that

$$\Delta \;=\; G \;+\; \text{readout(base)}, \tag{1}$$

where the readout term depends, to a good approximation, only on the *baseline accuracy* of the cell: it is negative below a crossing located at ∼32–40% accuracy, near zero at the crossing, and small and positive above it. The sign prediction holds in 79% of the 511 cells where measurement uncertainty allows it to be tested at all, and in 88% of those below the crossing (Section 5), and its quantitative form survived our hardest pre-registered tests: it predicted the feature gain of a backbone family it had never seen (Swin) from baseline heights alone, and at ImageNet scale it tied end-to-end gains to feature gains within 1.1 accuracy points on five of six backbone pairs (Section 7).

The law converts the grid into an explanation, and it is where the currencies stop being a metaphor and become the term G. Each intervention's is legible in what it does to frozen features: heavy augmentation supplies nuisance-invariance, contrastive pre-training supplies much the same invariance at greater expense, ImageNet weights supply mature photographic features, and the moment prior supplies oriented-energy structure. Same-currency fusions substitute; the prior and SimCLR each render the other nearly worthless, and their combination never beats the better single. Different-currency fusions stack; the prior's gain under DeiT-strength augmentation does not shrink but grows by a factor of 1.4–2.4. And fusing a shaping prior into an initialization that already carries mature features taxes accuracy in proportion to what the initialization supplied, a cost we show lands almost entirely on the features themselves rather than on the classifier (Section 6).

Two headline findings illustrate what the instrument can resolve. First, small vision transformers carry a large feature deficit that the prior fills: the gain reaches +13.0 points for ViT-S/16 and +26.0 for ViT-B/16 on ImageNet-100 at 224 px under the standard DeiT recipe, *growing* with model scale, and persists at +3.2 points after 1.28 million training images, where the ResNet baseline we measure is exactly neutral (+0.04). Second, the fashionable remedy is not always the right one: under plain augmentation SimCLR beats the prior across the mid-data band on photographic datasets, but under the modern DeiT recipe the comparison flips at every fraction we measure at matched ∼2× cost, because the recipe itself supplies the invariance SimCLR was being paid that compute to learn. Buying the comparator 5× instead removes the flip on Tiny-ImageNet and leaves it only above ∼10% of CIFAR-100 (Section 8), so the comparison is a statement about budget and population, not a ranking of methods.

Contributions.

1. **An instrument, not just experiments** (Section 3): a frozen recipe with fixed subsets, pinned filter banks, declared cost normalization, and pre-registered predictions with falsifiers, every headline table regenerable by one command from released artifacts.
2. **The grid** (Section 4): 2,978 configurations, 9,079 runs, comparing a free spectral prior against SSL, transfer, augmentation and learned teachers across 13 datasets, 9 backbones and 500 to 1.28M images.
3. **The law** (Section 5): $\Delta = G + \text{readout(base)}$, validated from CIFAR scale to ImageNet scale, with its boundary cases reported as findings.
4. **A fusion taxonomy with mechanism** (Section 6): stack, substitute, or tax, decided by whether the fused sources carry the same information currency, and verified on the feature side, not just end-to-end.
5. **A decision guide** (Section 12): a regime-indexed recommendation table distilled from the grid, with honest costs attached.

In one line, what the grid says: measure each source's currency alone and the result of fusing them is known in advance (Section 6); never pour a shaping prior into a strong initialization, because the tax scales with what that initialization was worth (Section 6.3); modern augmentation does not replace structure but amplifies it (Section 6.1); attention's feature deficit grows with model scale rather than shrinking (Section 7); and a comparator's budget is part of the claim, since two of ours did not survive raising it (Section 8).

2. Related work

We group prior work by the mechanism through which it supplies information to a data-limited network, since that is the axis along which our comparison is organized: hand-crafted structure placed in the forward path; auxiliary and distillation targets that shape features during training; self-supervised and transfer initializations that pay compute for representation; augmentation recipes that supply invariance; small-data results specific to attention; and the large-scale comparative benchmarks that most closely resemble what we build. Each paragraph below takes one of these in turn. Section 2.1 then places our three outcomes against the fusion literature's own account of how sources relate, and Section 2.2 states, in a single table, which properties a controlled fusion study requires and which of them each line of work currently provides.

Hand-crafted structure inside learned networks. Fixed oriented filters predate deep learning as image descriptors [1]. Scattering networks showed that cascades of fixed wavelets rival learned features in small-data regimes [2], and Gabor-structured convolutions have been used to regularize or cheapen early layers [3]. These lines place the structure *in the forward path*, where it occupies input bandwidth permanently. Our forward-path baselines reproduce their known failure mode, a mid-data penalty band that no filter choice escapes (Section 11), and our central object differs precisely in that the structure lives in a training-only auxiliary target whose influence decays to zero, so abundant data can override the prior rather than pay for it.

Auxiliary targets, deep supervision, and distillation. Regressing intermediate features onto targets is the mechanism of FitNets [4] and deep supervision [5]; HOG targets power MaskFeat's self-supervised pre-training [6]. What distinguishes our setting is the *fixedness and cost* of the target: no teacher network is trained, no pre-training stage is run, and the target is a closed-form function of the input. Our controls quantify how much this matters: a learned same-data teacher moves accuracy by at most 0.7 points anywhere on an eight-point envelope, HOG targets recover roughly half the moment target's gain, and a random fixed target does nothing (Table 20), so the gain is attributable to the moment structure itself, not to auxiliary regression as a mechanism.

Self-supervised learning (SSL) as initialization. Contrastive and self-distilling pre-training [7, 8, 9] is the modern default source of "free" features, and cookbook-level guidance exists for large-scale use. Much less is measured about the small-data, from-scratch regime we target. Our grid contributes three facts to that gap: negative-free SimSiam learns essentially nothing at 2.5–25k images under its published CIFAR recipe, though four times that budget recovers up to +4.3 and we report it as a property of the budget rather than of the method; SimCLR is strong on photographic statistics but loses to a free domain-agnostic prior on satellite imagery in the low-data band; and the SSL-versus-prior comparison inverts under a modern supervised recipe at matched cost, because the augmentation stack substitutes for the invariance SSL sells (Sections 6 and 8).

Transfer learning. ImageNet initialization remains the strongest single intervention on photo-like data, with known caveats under domain shift and with training-from-scratch competitive given enough data [10]. We use transfer both as a comparator and as a test of fusion: injecting the spectral prior *on top of* ImageNet weights is the one configuration in our grid that is destructive everywhere, and linear probing shows the destruction is feature-side, early shaping erases exactly the mature structure the initialization supplied, in proportion to how much that structure was worth (Section 6).

Small-data vision transformers. That ViTs underperform without large data or heavy regularization is well documented [11, 12], and a family of remedies exists: distillation tokens [12]; architectures that reintroduce the hierarchical, local inductive bias attention lacks, such as Swin [13], which we include as a backbone for that reason; and auxiliary self-supervised objectives trained jointly with the classifier, of which Liu et al. [14] is the closest antecedent to this work — a dense relative-localization task added for exactly this deficit, differing from ours in that its target is computed from the images themselves rather than pinned in advance. Our contribution to this literature is quantitative and comparative: we measure the deficit as a feature quantity ($G \approx$ 13–15 points for ViT-tiny across 2.5–12.5k images, roughly 2.3× any convolutional backbone), show a fixed spectral target fills it more cheaply than contrastive pre-training and more effectively than DeiT augmentation alone, and show the deficit *grows* with model scale at fixed data (Section 7), which argues it reflects data-hunger rather than small-model capacity.

Large-scale comparative benchmarks. Two recent efforts share our comparative ambition and clarify what is still missing. *Battle of the Backbones* [15] compares a broad suite of pretrained backbones across tasks over more than 1,500 training runs, and finds supervised convolutional pre-training still competitive with self-supervised and vision-language alternatives. Marks et al. [16] take a closer look at self-supervised benchmarking specifically, and show that conclusions depend heavily on the evaluation protocol chosen. Both benchmark *pretrained backbones*, so their unit of comparison is a checkpoint someone else produced, and neither normalizes by training cost nor asks what happens when two sources of information are combined. Our unit is instead the *intervention applied to a fixed recipe*, which is what makes cost normalization and the stacking question answerable at all.

Spectral structure inside transformers. A parallel line injects spectral structure into attention architectures: a scattering-based transformer [17] performs spectral decomposition with dual-tree complex wavelets inside the token mixer, and FViT [18] adds a *learnable* Gabor filter to focus attention across scales and orientations. These are architectural changes that persist at inference, and each is evaluated as a model proposal against published baselines. Our object is the complement: the filters are fixed rather than learned, they never enter the forward path, and the deployed network is unchanged, which is what lets us ask the benchmark question of when adding them pays rather than the model question of whether one architecture beats another.

2.1. Relation to fusion theory

The outcomes we measure are not new categories, and we set out the prior claims before our own. Durrant-Whyte [19] classified the interactions between disparate information sources as *cooperative*, *competitive* and *complementary*; the surveyed form now standard in the field [20] states the triple as complementary, redundant and cooperative, where complementary sources describe different aspects of the target, redundant sources describe the same aspect, and cooperative sources jointly yield what neither yields alone. Our *stack* is complementary combination and our *substitute* is redundant combination. We adopt that vocabulary here and use our own terms only as informal labels for it.

Dasarathy [21] organizes the field's questions as what, where, why, when and how to fuse, and observes that the *when* has received least attention because it is the one that cannot be answered from the architecture. That is the question this paper takes up, in the specific setting where one source is hand-crafted and the other is learned from the data. Nor is knowledge fusion itself a new object: Smirnov and Levashova [22] surveys the patterns by which knowledge and data are combined, and catalogs the same three outcomes we recover. What that survey records as patterns to be chosen by a designer, we measure as outcomes to be predicted from the sources, which is the difference between a taxonomy and a decision procedure. The field's benchmarks have generally compared *algorithms* for a fixed fusion problem; ours compares *sources* under a fixed algorithm, holding the recipe frozen so that what varies is the knowledge injected rather than the method injecting it.

One difference is substantive rather than terminological, and it sharpens what a representation-learning setting adds. In sensor fusion redundancy is a *virtue*: two instruments measuring the same quantity reduce variance and buy fault tolerance. In ours it is waste, because performance is bounded by the information the sources share rather than improved by supplying it twice. Redundancy pays for reliability and not for representation quality, and a taxonomy built for the first case does not transfer its sign to the second.

Our third outcome, in which a source destroys what its partner supplied, is likewise not new, and has at least three names. It is *catastrophic fusion* in the sense of Movellan and Mineiro [23], where modules combined under mismatched assumptions do worse than the better module alone; it is a form of negative transfer [24]; and on the multimodal side it is the modality competition of Wang et al. [25], who report that the best unimodal network often beats the jointly trained multimodal one despite the latter receiving strictly more information. What we add is not the phenomenon but its magnitude law: the damage scales with how much the displaced source had contributed, and vanishes exactly where it had contributed nothing.

The formal account of our informal "currency" is partial information decomposition [26], which splits what two sources tell us about a target into unique, redundant and synergistic atoms, and which Liang et al. [27] have already carried into deep multimodal learning with estimators that scale and with a-priori model selection as the application. We do not compute it. Our measurement is a linear-evaluation gap on frozen features: cheap enough for 2,978 cells, ordinal rather than an information quantity, and blind to anything not linearly decodable. Read our result as an empirical instance of the decomposition those authors formalize, not as an estimate of it.

Two further connections position the predictor rather than the taxonomy. Predicting a downstream outcome from a cheap measurement, without training the combined system, is the programme of transferability estimation [28], which scores a single pre-trained source; ours asks the same question of a *pair*. And the challenge that programme raises for us is direct: Newell and Deng [29] report that linear evaluation does not correlate with fine-tuning performance in general. Our defence is narrow and empirical rather than a rebuttal. We do not use the linear evaluation to rank absolute quality; we use the *gap* between two arms probed under

Table 1
Where the literature stands on the five properties a fusion question needs, and what this work adds. ✓ = property present; ○ = partial or incidental; ✗ = absent. "Cost-normalized" means comparators are reported against a declared training-compute multiple rather than at whatever budget each method's authors chose; "feature-side" means the effect is measured on frozen representations, not only end-to-end; "combination" means the study measures what happens when two sources are applied together.

Line of work	one frozen recipe	cost-normalized	feature-side	combinations	predictive model
Fixed spectral filters in the forward path [2, 3, 18]	○	✗	✗	✗	✗
Auxiliary and distillation targets [4, 5, 6]	○	✗	○	✗	✗
Self-supervised pre-training [7, 8, 9]	✗	✗	✓	✗	✗
Transfer learning studies [10]	○	✗	○	✗	✗
Small-data ViT recipes [12]	✗	○	✗	○	✗
Backbone and self-supervised benchmarks [15, 16]	○	✗	✓	✗	✗
Classical fusion taxonomy [19, 26]	✗	✗	✗	✓	○
This work	✓	✓	✓	✓	✓

an identical protocol, on checkpoints that differ in one intervention, and we report how well that gap predicts the paired difference (Section 5) rather than assuming it does.

Finally, our knowledge source sits where von Rueden et al. [30] make precise. Their taxonomy classifies informed learning by knowledge *source*, *representation* and *integration point*, and states that informed learning requires two independent sources of information, data and prior knowledge, which is the reading of "two sources" under which this study is a fusion study at all. Ours is scientific knowledge about early vision, represented as a pinned filter bank, integrated at the learning algorithm; the nearest existing method is their own informed pre-training, which condenses knowledge into prototypes for initialization rather than into a decaying auxiliary target. That survey catalogs where knowledge can be injected. What it does not settle, and what a practitioner must decide, is *when* injecting it is worth anything given what the data and the available pre-trained artifacts already supply. That is the question this benchmark is built to answer, and we are careful to claim only that: we found no controlled, compute-declared answer to it, not that no neighbouring question has been answered.

2.2. The gap this paper closes

Table 1 summarizes the position. Each column is a property that a question about *fusing* information sources requires, and no existing line supplies all five at once.

The first column matters because comparisons made under each method's own preferred recipe cannot attribute a difference to the method. Benchmarks of pretrained backbones [15, 16] inherit whatever recipe produced each checkpoint, and Marks et al. [16] show directly that the evaluation protocol can change the conclusion. We instead hold one recipe and one set of fixed image indices fixed and vary only the intervention.

The second is rarely reported at all. Self-supervised pre-training costs roughly twice the training compute of the baseline it improves, but that factor almost never appears next to the accuracy it buys, which makes "method A beats method B" unanswerable for a practitioner with a fixed budget. Every comparator here carries a declared multiple.

The third and fourth columns are what turn a table of numbers into an explanation. Linear probing is standard practice for evaluating self-supervised representations, so the feature-side view exists in that literature; what is missing is applying it to a *paired* comparison so the same intervention is measured end-to-end and on frozen features at once, which is what exposes redundancy between two sources. Combinations are the least studied of all: the question of what happens when a prior is applied on top of augmentation, on top of self-supervised weights, or on top of a pretrained initialization is precisely the fusion question, and we could find no controlled study that measures all three.

The fifth column is the one we regard as the paper's distinguishing contribution. Benchmarks describe; a law predicts. Our decomposition $\Delta = G + \text{readout(base)}$ was registered in advance and used to call the feature gain of a backbone family it had never seen, of four unseen domains, and of ImageNet-scale cells, before those measurements existed (Sections 5 and 7). A related decomposition of learning dynamics into representation and readout components has been proposed for grokking and double descent; to our knowledge no comparable predictive account exists for data-efficiency interventions.

Benchmark methodology. Reality-check studies have repeatedly shown that uncontrolled recipes manufacture illusory progress, and reproducibility in machine learning remains fragile. We adopt the strictest discipline we could operationalize: a single frozen recipe; fixed per-subset indices so every intervention sees byte-identical images; numerically fingerprinted filter banks; deviations quarantined marked as such and never mixed into headline tables; and, unusual in this literature, predictions with explicit falsifiers registered *before* each wave of runs, with misses reported as plainly as hits (Sections 3, 7).

3. The instrument

The benchmark's claims rest on the comparisons being controlled, so we describe the controls before the results (Fig. 1). This section is organized as the instrument's six components, in the order a reader would need to trust a single number. Section 3.1 fixes the training recipe and the fixed data subsets that

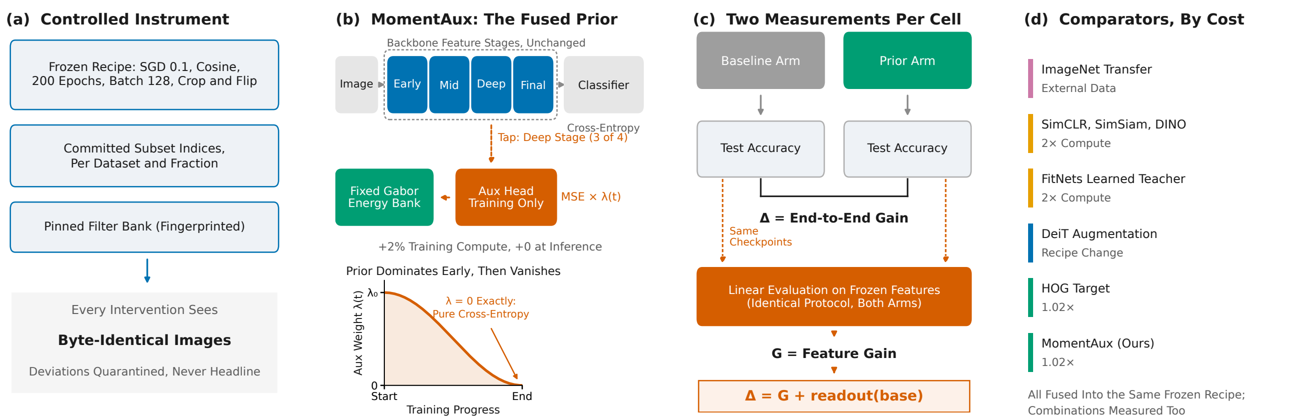

Figure 1: The study in one picture. **(a)** One frozen recipe and fixed subset indices, so every intervention is measured on byte-identical images; cells that must deviate are marked diagnostic and never enter headline tables. **(b)** MomentAux, the fused knowledge source: a pinned bank of oriented-energy targets regressed from a tap at the third of the backbone's four feature stages, during training only, with the loss weight λ decaying to exactly zero, so the deployed network is byte-identical to the baseline and high-data neutrality is structural. **(c)** Each cell yields two measurements, the end-to-end gain Δ and the frozen-feature gain G from an identically-configured linear evaluation, whose difference defines the readout term of Eq. 1. **(d)** The comparator ladder, ordered by declared training cost; all are fused into the same frozen recipe, and their pairwise combinations are measured as well.

make any two cells comparable. Section 3.2 specifies the knowledge source under study, the pinned spectral bank and the auxiliary head that injects it. Section 3.3 defines the competing interventions and the cost normalization under which they are compared. Section 3.4 introduces the linear evaluation of frozen features that supplies the quantity G, without which the law of Section 5 could not be measured at all. Section 3.5 states the seed and uncertainty protocol, Sections 3.6 and 3.7 disclose how the configuration under test was selected and how that leaves it advantaged relative to the comparators, and Section 3.8 describes the pre-registration discipline that turns the grid's predictions into falsifiable ones.

3.1. Frozen recipe and fixed subsets

Every headline cell trains with an identical recipe: stochastic gradient descent (SGD) with momentum 0.9, learning rate 0.1, weight decay 5×10^{-4}, cosine schedule, 200 epochs, batch size 128, and random-crop + horizontal-flip augmentation only. The recipe is frozen in the strict sense: no cell tunes it. Configurations that must deviate, AdamW [31] for architectures the SGD recipe cannot train, reduced epochs at ImageNet scale, added augmentation stacks, are marked as diagnostic in their configuration name, which the training code enforces, and never enter a headline table. Within such cells both arms share the deviation, so the paired difference remains valid even though the absolute numbers are not comparable to frozen-recipe cells.

Data fractions are not sampled at run time. For each dataset $\mathcal{D}$ and fraction p we precompute a stratified index set $S_{\mathcal{D},p}$ by drawing, per class, a seeded permutation truncated to $p\%$, and fix it once for the whole study. Every intervention therefore consumes byte-identical images, which is what makes an accuracy difference attributable to the intervention rather than to the draw. Because the recipe fixes epochs rather than steps, the optimization budget scales with $|S_{\mathcal{D},p}|$; we treat data and compute as jointly varied along this axis and say so wherever it matters.

Two further controls proved necessary in practice. Dataloader worker count is part of the reproducibility contract: PyTorch seeds each worker's augmentation RNG from a base seed plus worker index, so changing the count re-draws the augmentation stream. The change is unbiased, it acts like a different augmentation seed, but it breaks byte-level reproducibility, so the count is pinned per cell and recorded per run. Second, the filter bank is *pinned*: unit tests fingerprint its numerical values, so no measurement can silently drift with a library version.

3.2. The prior: MomentAux

The knowledge source. The bank consists of $N=8$ complex Gabor quadrature pairs, two spatial scales $\sigma \in \{2,3\}$ crossed with four orientations $\theta \in \{0,\pi/4,\pi/2,3\pi/4\}$; each of the form

$$g_{\sigma,\theta}(\mathbf{u}) = e^{-\|\mathbf{u}\|^2/2\sigma^2}\, e^{\,i\,k_\sigma\,\mathbf{u}^\top(\cos\theta,\sin\theta)}, \qquad (2)$$

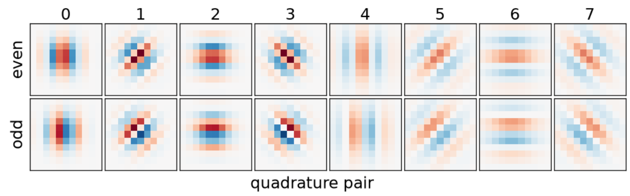

Figure 2: The knowledge source, in full. The eight complex Gabor quadrature pairs of Eq. 2: two scales crossed with four orientations, even (top) and odd (bottom) components of each pair. The auxiliary target is the pointwise magnitude of the pair responses (Eq. 3), which is what makes it phase-invariant. The bank is fixed before any training, identical for every dataset, backbone and data fraction in the study, and fingerprinted by a regression test so that no result silently changes bank.

with k_σ set so each envelope carries a comparable number of cycles (Fig. 2). The target is the *phase-invariant magnitude* of the response to the greyscale input x,

$$m_{\sigma,\theta}(x)[u] \;=\; \big|\,(x * g_{\sigma,\theta})[u]\,\big|, \tag{3}$$

taken pointwise at every spatial location u, then stacked over the N pairs and average-pooled to the tapped stage's spatial resolution, giving $\mathbf{m}(x) \in \mathbb{R}^{N\times h\times w}$. Magnitude rather than raw filter response is the operative choice: the ablation of Section 11 shows oriented *edges* are a poor target (-0.24) while their phase-invariant energy is the best one measured ($+2.81$ at the same cell).

The fusion mechanism. Let $f_3(x)$ be the activations of the backbone's third feature stage, the third of four in every architecture we use (`layer3` in a ResNet, `stages.2` in ConvNeXt, the matching block group in ViT and Swin), and W a 1×1 convolution mapping them to N channels. Training minimizes

$$\begin{aligned}\mathcal{L}(t) \;&=\; \mathcal{L}_{\mathrm{CE}} \;+\; \lambda(t)\,\big\|\,W f_3(x) - \mathbf{m}(x)\,\big\|_2^2,\\ \lambda(t) \;&=\; \tfrac{\lambda_0}{2}\Big(1+\cos\tfrac{\pi t}{T}\Big),\end{aligned} \tag{4}$$

so the prior dominates early and vanishes by the final epoch. Three properties follow directly and each is load-bearing for the paper's claims. (i) Because $\lambda(T) = 0$ *exactly*, late training is pure cross-entropy and neutrality at sufficiency is structural, not tuned, the right-hand end of every envelope in Table 4 is a prediction of the schedule, not a fitted result. (ii) W is discarded after training, so the deployed network is byte-identical to the baseline: same parameters, same floating-point operations (FLOPs), zero inference cost. (iii) The measured training overhead is $\sim$2%, against 100% for a self-supervised pre-training stage.

The scale degeneracy and its fix. The auxiliary objective in Eq. 4 is invariant under replacing f_3 by f_3/c and W by cW, which lets SGD satisfy it by collapsing the tapped features and inflating the head rather than by learning structure. We traced this directly: on ResNet-50 the tapped standard deviation fell from 0.596 to 0.051 while $\|W\|$ grew from 1.64 to 11.2 and cross-entropy stalled at chance. Projecting $\|W\|$ back to its initialization norm after each step removes the degenerate direction, converting a bistable failure into a stable +3.9 (Section 11); we adopt it always-on. One configuration, λ_0=1.0, magnitude target, third-stage tap, head-norm on, transplants across every dataset and backbone without retuning, and we use it verbatim as the *champion* unless stated.

3.3. Comparators and cost normalization

We normalize by declared training cost relative to the baseline: MomentAux $\approx$1.02×; SimCLR, SimSiam and DINO initializations 2× (a full pre-training stage, then the frozen recipe); DeiT-strength augmentation [12] $\approx$1× compute but a modern recipe change; FitNets teachers 2×; HOG targets $\approx$1.02×; ImageNet transfer amortized, but flagged throughout as the one comparator with an external-data advantage.

Every self-supervised stage runs on the same subset images of the cell it initializes, never on extra data, for 200 epochs at batch 128 with that method's own published view augmentations, and is followed by the unchanged frozen recipe. Because a weak comparator would flatter our result, we test SimCLR's configuration explicitly rather than assume it: strengthening its contrastive views to DeiT strength makes it *worse* at every fraction measured (Section 6), so its standard views are its best shot at this scale.

3.4. Measuring features: linear evaluation

The law needs a feature-side quantity, and we obtain it by *linear evaluation* of frozen representations, the standard linear-probe protocol [32]. For each trained checkpoint we freeze the network, extract penultimate features $z = \mathrm{pool}(f_L(x))$ for the full training split under the deterministic (no-augmentation) transform, and fit a multinomial logistic regression using the limited-memory BFGS optimizer (L-BFGS) and fixed hyper-parameters:

$$\min_V \;\tfrac{1}{n}\sum_i \mathrm{CE}\big(V z_i, y_i\big) \;+\; 10^{-4}\|V\|_2^2. \tag{5}$$

Only the features differ between arms; everything else is identical. We define the feature gain as the linear-evaluation gap $G = \mathrm{acc}_{\mathrm{lin}}^{\mathrm{aux}} - \mathrm{acc}_{\mathrm{lin}}^{\mathrm{base}}$, and the readout term as the residual $\Delta - G$.

Linear evaluations are diagnostics and never headline numbers; they read labels the cell never saw. Two scope rules, both learned the hard way and enforced throughout: an evaluation is interpretable only while it holds substantially more labels than the cell trained on, so at 100% cells we report the aux-vs-baseline gap but do not split it into G and readout; and at

ImageNet64 scale, where an L-BFGS fit over 1.28M rows is impractical, evaluations use fixed per-class budgets whose G values are compared only to other same-budget evaluations.

3.5. Statistical protocol

Each cell runs ≥ 3 seeds, ten for headline envelopes. For a paired cell we report Δ as the difference of arm means with the standard error of the mean (SEM) of that difference.

The readout term needs more care than a difference of two independent measurements, because Δ and G are not independent: the linear evaluation probes the very checkpoints whose end-to-end accuracy gives Δ, so across seeds the two move together and $\mathrm{Cov}(\Delta, G) > 0$. Propagating the residual as $\sqrt{\mathrm{SEM}(\Delta)^2 + \mathrm{SEM}(G)^2}$ would therefore *overstate* its uncertainty; on this grid it does so by a median factor of 1.8. We instead form the readout *per seed* and take the standard error of that quantity directly,

$$\mathrm{readout}_s = \left(a_s^{\mathrm{aux}} - e_s^{\mathrm{aux}}\right) - \left(a_s^{\mathrm{base}} - e_s^{\mathrm{base}}\right), \quad (6)$$

where a_s and e_s are seed s's end-to-end and linear-evaluation accuracies, and only seeds present in all four arms contribute. A cell is called *resolvable* when $|\mathrm{readout}| > 2\,\mathrm{SEM}$; only resolvable cells can test the sign law, and Section 5 explains why counting the rest would be misleading rather than conservative. Cells in which any seed collapses to chance are flagged *bistable*, excluded from headline status, and reported as findings about trainability in their own right.

Four properties of this protocol are worth stating plainly, because each bounds what the audit of Section 5.2 can claim.

The two-SEM rule is a declared magnitude threshold, not an exact test. With three seeds per arm the Student-t multiplier for a 95% interval exceeds two, so the criterion is mildly liberal at the minimum seed count. We therefore fix it in advance and report the audit across a range of thresholds rather than defending one value (Table 8). Because the seed-paired standard error is smaller than an independent propagation would give, this rule admits about 1.8 times as many cells to the audit as the naive formula does, and the extra cells are the borderline ones. That makes the test harder, not easier.

Cells are not independent. A baseline arm is shared by every variant paired against it, and cells recur across the same datasets and backbones, so the 511 resolvable cells come from 258 distinct (dataset, backbone, fraction) groups. We therefore report a cluster-level audit alongside the cell-level one, and treat exact p-values as indicative rather than literal.

The aggregate claim is not a family of per-cell tests. We never assert significance for an individual cell, so no multiplicity correction applies to the hit rate; the resolvability filter is a selection rule applied identically to every cell, declared before the audit.

Three-seed deltas are slightly optimistic. When we deepened the three headline CIFAR-100 cells from three to ten seeds per arm, every Δ shrank, by 0.07, 0.15 and 0.39 points. The direction reproduces an earlier CIFAR-10 deepening. Nothing in the shape of any envelope changed, but a three-seed Δ should be read as an upper-ish estimate, and headline envelopes use ten seeds for that reason.

3.6. Configuration selection, and what it costs us

The champion configuration was not given; it was chosen. Target family, tap depth, λ_0, head-norm and loss form were each swept, and those sweeps were run on CIFAR-100 and scored on the same test split that supplies this paper's CIFAR-100 numbers. We did not hold out a validation split, and we state the consequence rather than leave it to be inferred: *the CIFAR-100 cells are a selection set, and their absolute values carry a selection bias of unknown size.*

Two things limit the damage, and one argument survives intact. The sweeps selected among a small number of pre-declared alternatives rather than searching a continuous space, and the chosen setting was fixed once and then applied verbatim everywhere else. That is what makes the transplant evidence the load-bearing evidence here: the same configuration, with no retuning, was applied to eleven further datasets, nine backbones and two ImageNet-scale stages that played no part in its selection, and its behavior there is not selection-contaminated. Where this paper makes a general claim, it should be read as resting on those populations, with CIFAR-100 as the development set that produced the candidate.

We considered re-selecting the configuration on a held-out validation split and decided against it, and state that here rather than leave a reader wondering. Re-selection would invalidate every cell in the grid, since the selected configuration is the one 2,978 cells were trained with, so the honest choices were to disclose the contamination or to retrain the study. Given that the general claims already rest on eleven populations that took no part in selection, we judged disclosure to carry the information a reader needs at a fraction of the cost. A future benchmark built on this harness should split a validation fold from the start; ours did not, and that is a design defect rather than a defensible choice.

3.7. Comparator parity

Cost normalization fixes the comparators' budget, and that cuts both ways. Every comparator is run at its published defaults inside the frozen recipe, at the declared multiple of the baseline's training cost, on the cell's own subset. The prior, by contrast, was developed here and therefore received the sweep just described. We did not sweep the comparators.

The sharpest consequence is at the smallest fractions. Self-supervised pre-training at 1% of CIFAR-100 means 200 epochs of contrastive learning over 500 images, which is far short of the schedules used in published small-data practice, and our check that strengthening SimCLR's views does not help (Section 3.3) varies the view distribution, not the budget. So a statement such as "SimSiam learns almost nothing at this scale" is, strictly, a statement about SimSiam *at this budget*; undertraining and method weakness are not separable under a cost-normalized protocol, by construction. Readers who care about the best attainable accuracy rather than the best accuracy per unit compute should treat our self-supervised numbers as lower bounds.

Rather than leave that as a caveat, we measured it. Section 8 re-runs both self-supervised comparators at four times their pre-training budget across the data envelope and reports what changes, including the two claims of ours that do not survive it.

3.8. Pre-registration

Each experimental wave was launched with numeric prediction bands and explicit falsifiers recorded in a project ledger before any result existed, and outcomes are reported against those bands including the misses. Several of our predictions failed instructively, "SSL is data-regime-bounded" (wrong: SSL won at every conv fraction under plain augmentation) and "augmentation substitutes for the prior" (wrong in the opposite direction: it amplifies), and both failures became findings in Section 6. We regard this discipline as part of the instrument rather than a presentational choice: a benchmark whose predictions cannot fail is not measuring anything.

4. The grid

What a cell is. We report the grid in *cells*, and the term carries a specific meaning throughout. A cell is one configuration, identified by its configuration name, aggregated over the seeds trained under it: a dataset, a data fraction, a backbone, and one intervention with its hyper-parameters fixed. Every number we quote for a cell is a mean over its seeds, and every uncertainty is the standard error of that mean.

Two consequences are worth stating because they affect how the counts should be read. First, cells are grouped by configuration name and never by parsed fields, since two configurations can agree on dataset, backbone and intervention while differing in a filter-bank argument or a calibration setting; grouping on fields would silently average different variants together. Second, a cell is not a run. The 2,978 cells reported here rest on **9,079 individual training runs**, and the pairing that gives every Δ its meaning is between cells that share a dataset, fraction and backbone and differ in exactly one intervention.

Table 2
The benchmark at a glance. Every cell trains the frozen recipe of Section 3.1; costs are declared multiples of baseline training compute.

Axis	Coverage
Datasets	CIFAR-10, CIFAR-100, ciFAIR-100, STL-10, Tiny-ImageNet, coarse variants; EuroSAT, DTD, Food-101, PathMNIST, CUB-200; ImageNet64, ImageNet-100
Domains	photographic, satellite, texture, food, histopathology
Backbones	ResNet-18, -34 and -50 [33], MobileNetV3 [34], ConvNeXt-T [35], ViT-tiny, -S and -B [11], Swin-T [13]
Data scale	500 to 1.28M images; 10 to 1000 classes
Resolution	32, 64, 96, 224 px
Interventions	MomentAux ($1.02\times$); SimCLR, SimSiam and DINO ($2\times$); ImageNet transfer; DeiT augmentation; FitNets teacher ($2\times$); HOG target; pairwise combinations
Measured	2,978 cells over 9,079 retained run records including exploration and repeats

The count includes every cell we trained, diagnostics and ablations included, not only those that appear in a headline table; cells that deviate from the frozen recipe are named so in their configuration and are never mixed into headline comparisons (Section 3.1).

Table 2 summarizes the design space, and the rest of this section unpacks it in five steps. Section 4.1 describes the 13 datasets and the six domains they span, and Section 4.2 reports the hardware, wall-clock and carbon budget the grid consumed. The remaining three subsections are the three views of the grid that the results are built from: Section 4.3 gives the prior's own gain envelope across data fractions, which is the object the law of Section 5 explains; Section 4.4 places the competing interventions on the same envelope at matched cost, which is what Section 6 analyzes; and Section 4.5 isolates the attention backbones, where the effect is largest and where the scale validation of Section 7 concentrates. Two table conventions hold throughout and are stated here once rather than repeated in every caption: alternate rows are lightly shaded for readability, and ↑ (↓) marks any column in which a higher (lower) value is the better outcome. Columns holding a signed method-versus-method difference carry no marker, because neither sign of such a difference is better a priori. A dash marks a cell we did not measure.

4.1. Datasets

Table 3 lists the datasets. The selection is deliberate rather than convenient. Five visual domains are represented because the currency account predicts that which information source is scarce depends on image statistics, and a benchmark drawn only from photographic data could not test that: satellite imagery has non-photographic color statistics and meaningful rotations, texture lacks object-level structure, histopathology is stain-dominated, and fine-grained birds require

Table 3
Datasets and the partitions they were trained on. Counts at each fraction are read from the fixed subset index files that every arm of every cell consumes, not recomputed from the percentage, so they include the rounding the per-class stratified draw introduces. Parentheses give images per class, which is what decides a cell's regime: 1% of CIFAR-10 is 50 images per class while 1% of Tiny-ImageNet is 5, and the prior behaves very differently at those two points. Dashes are fractions not run for that dataset; the two ImageNet stages were run at full data only. The relabeled controls (CIFAR-100-super, Tiny-ImageNet-20, -20b, -super, -semantic) and the multi-source views (EuroSAT-MS, So2Sat SAR/optical/both) reuse their parent's indices exactly, so they share its row. ciFAIR-100 [36] is not listed: it replaces the CIFAR-100 *test* set only, as a check that low-data gains are not memorized train/test duplicates, and has no training partition. Two kinds of blank are distinguished: $\times$ marks a fraction that *cannot* be trained under the frozen recipe, because it yields fewer than one batch of 128 images and the loader is empty (the reason no dataset smaller than DTD has a 1% cell); a dash marks a fraction that is runnable but that we did not run.

				training images at fraction (per class)										
Domain	Dataset	Cls.	Px	1%	2%	3%	5%	7%	10%	15%	20%	25%	50%	100%
Photographic	CIFAR-10 [37]	10	32	500 (50)	1,000 (100)	1,500 (150)	2,500 (250)	3,500 (350)	5,000 (500)	7,500 (750)	10,000 (1000)	12,500 (1250)	25,000 (2500)	50,000 (5000)
	CIFAR-100 [37]	100	32	500 (5)	1,000 (10)	1,500 (15)	2,500 (25)	3,500 (35)	5,000 (50)	7,500 (75)	10,000 (100)	12,500 (125)	25,000 (250)	50,000 (500)
	STL-10 [38]	10	96	×	×	150 (15)	250 (25)	350 (35)	500 (50)	750 (75)	1,000 (100)	1,250 (125)	2,500 (250)	5,000 (500)
	Tiny-ImageNet [39]	200	64	1,000 (5)	2,000 (10)	3,000 (15)	5,000 (25)	7,000 (35)	10,000 (50)	15,000 (75)	20,000 (100)	25,000 (125)	50,000 (250)	100,000 (500)
Satellite	EuroSAT [40]	10	64	216 (22)	432 (43)	648 (65)	1,080 (108)	1,512 (151)	2,160 (216)	3,240 (324)	4,320 (432)	5,400 (540)	10,800 (1080)	21,600 (2160)
Multi-sensor	So2Sat LCZ42 [41]	17	32	243 (14)	481 (28)	–	1,206 (71)	–	2,412 (142)	–	–	6,028 (355)	–	24,119 (1419)
Texture	DTD [42]	47	64	×	×	×	188 (4)	282 (6)	376 (8)	564 (12)	752 (16)	940 (20)	1,880 (40)	3,760 (80)
Food	Food-101 [43]	101	64	808 (8)	1,515 (15)	2,222 (22)	3,838 (38)	5,252 (52)	7,575 (75)	11,312 (112)	15,150 (150)	18,988 (188)	37,875 (375)	75,750 (750)
Histopathology	PathMNIST [44]	9	64	901 (100)	1,800 (200)	2,700 (300)	4,498 (500)	6,300 (700)	9,000 (1000)	13,499 (1500)	17,998 (2000)	22,500 (2500)	44,996 (5000)	89,996 (10000)
Fine-grained	CUB-200 [45]	200	64	×	×	200 (1)	394 (2)	400 (2)	600 (3)	800 (4)	1,200 (6)	1,594 (8)	2,994 (15)	5,994 (30)
ImageNet scale	ImageNet64 [46]	1000	64	–	–	–	–	–	–	–	–	–	–	1,281,167 (1281)
	ImageNet-100 [47]	100	224	–	–	–	–	–	–	–	–	–	–	126,689 (1267)

discriminating parts rather than categories. PathMNIST is a public, de-identified benchmark distributed as part of MedMNIST v2 [44]; its use here involves no new data collection and required no ethics approval. The two ImageNet stages buy the axes small datasets cannot: ImageNet64 supplies data and label scale at fixed resolution, while ImageNet-100 supplies native 224 px resolution and a model-scale curve.

Two training pools differ from the sizes usually quoted for those datasets, and Table 3 reports the pool we actually drew from rather than the headline split. For DTD we follow the standard protocol and train on train+val (3,760 images, 80 per class), testing on the held-out test split. For So2Sat we train on the v4 *validation* split (24,119) and test on the v4 test split: both are held out from the training cities, which makes this a harder generalization setting than the canonical split and is why the 352,366-image training split plays no part here.

A second group exists purely as controls. CIFAR-100-super relabels byte-identical CIFAR-100 subsets with 20 coarse classes, so class count changes while pixels and optimization steps do not; the Tiny-ImageNet variants do the same on a second population, including a semantically ordered grouping and an arbitrary positional one whose contrast isolates whether coherence or count is doing the work (Appendix C). ciFAIR-100 replaces the 927 CIFAR-100 test images that duplicate training images, which lets us verify that low-data gains are not memorized duplicates.

4.2. Implementation, compute and carbon

Training uses PyTorch with automatic mixed precision and `timm` model definitions, pinned per machine and recorded in every run record (PyTorch 2.4–2.8, `timm` 1.0.27, CUDA 12.x); metrics come from `torchmetrics` with a scikit-learn cross-check in the test suite rather than hand-rolled implementations. The study ran on three systems: the MareNostrum5 accelerated partition at the Barcelona Supercomputing Center, using up to 40 NVIDIA H100 GPUs concurrently (ten four-GPU nodes) under a shared work queue so that heterogeneous cell costs do not idle accelerators; a university cluster with H100 and H200 nodes; and two local NVIDIA RTX 3090 workstations used for evaluation passes and analysis. Because these small-image cells are dataloader-bound rather than compute-bound, we run two to three trainings per GPU, sized so that the heaviest cells still fit in memory.

Aggregating the wall-clock recorded in all 9,079 retained run records, the study consumed **3,518 GPU-hours**: 3,080 on H100, 82 on H100 NVL and

Table 4
Complete envelopes: gain Δ of the champion prior over the shared baseline (points). Bold: ten seeds per arm; others three. Per-cell SEMs (0.07–0.7) ship with the released records.

Fraction	CIFAR-100 ↑	CIFAR-10 ↑	Tiny-IN ↑
1%	**+1.42**	**+6.37**	+1.49
2%	+2.50	**+6.66**	+1.81
3%	+3.68	+5.38	–
5%	**+5.15**	+4.41	**+2.12**
7%	+4.87	+2.21	–
10%	**+3.75**	+1.09	+1.65
15%	+2.55	−0.66	–
25%	+0.25	−0.83	+0.10
100%	+0.08	−0.26	−0.42

Table 5
Domain datasets at representative fractions (points, three seeds). The unimodal shape recurs on satellite, texture, food, histopathology and fine-grained data. PathMNIST above $\sim$15% is confounded by a shifted test split that penalizes *all* methods (Appendix D).

Dataset	1% ↑	5% ↑	10–15% ↑	≥25% ↑
EuroSAT	+2.67	+0.84	+1.22	$\sim$0
PathMNIST	+5.70	–	+1.74	conf.
Food-101	–	+5.63	+3.18	−0.47
STL-10	–	–	+5.92	+3.18
DTD	–	+0.30	+2.15	+3.55*
CUB-200	–	–	–	+2.74*

*value at 100%.

H200 NVL, and 356 on RTX 3090. Taking device thermal design power as an upper bound on draw, a data-center power usage effectiveness of 1.2, and a Spanish grid carbon intensity of 0.17 kg CO_2eq per kWh, this corresponds to approximately 2,787 kWh and **474 kg CO_2eq**, about 52 g CO_2eq per training run. We report this as an order-of-magnitude accounting following the methodology of the Machine Learning Impact calculator [48], not a measurement. Its known omissions should be read with it: it assumes sustained peak GPU power and uniform throughput, ignores idle and host-side draw, excludes embodied (manufacturing) carbon and storage, and applies a single national grid intensity rather than the regional and hourly value at the time each job ran. The figure covers every run whose record we retain, which is every run whose cell survives in the released grid; runs superseded during the campaign leave no record, because the training entry point refuses to overwrite a completed cell and so never re-times one. Two aspects of the study's design lower this figure materially. The intervention under test adds only about 2% to a run, so the comparison itself is cheap; the expense lies in the comparators, since every self-supervised arm doubles the cost of its cell. And re-running an existing cell has no effect, which matters more than it sounds: several queue faults during the campaign would otherwise have silently recomputed finished work.

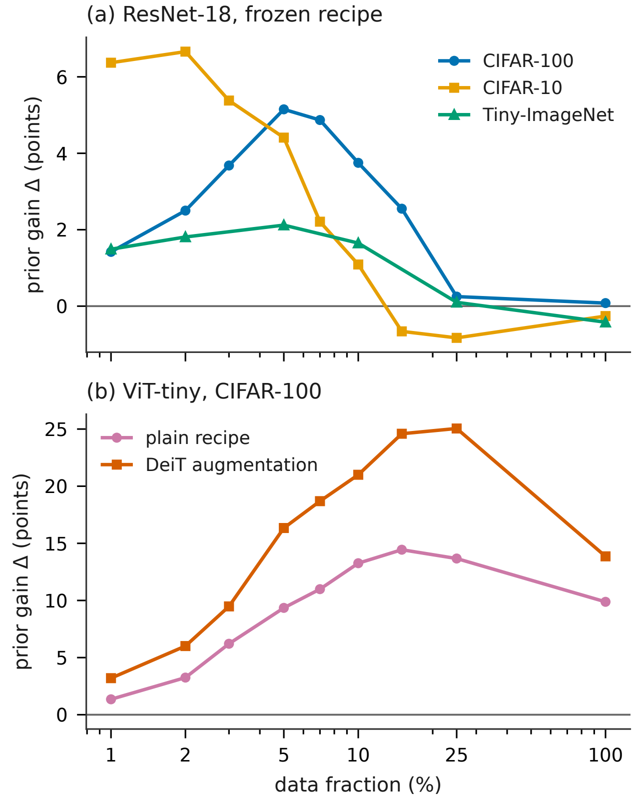

Figure 3: The prior's gain across the data axis. (a) On convolutional backbones the envelope is unimodal and neutral at sufficiency; the peak tracks task difficulty, not fraction. (b) On ViT-tiny the gain is large at *every* scale, and DeiT-strength augmentation amplifies rather than replaces it (Section 6).

4.3. The prior's own envelope

Tables 4–5 and Fig. 3a show the prior's gain across the data axis. Three regularities recur. The envelope is *unimodal*: gains rise from the extreme-scarcity floor, peak in a dataset-specific band, and decay toward zero as data becomes sufficient. The peak's location tracks the dataset's difficulty rather than its fraction, easy ten-way CIFAR-10 plateaus across 1–2% (the two cells differ by 0.29, which their seed uncertainty cannot resolve) and is already negative at 15%, while hundred-way CIFAR-100 peaks at 5%. And at full data the prior is neutral within noise on every one of ten populations, which is what the the decaying weight schedule is for: the fusion asks nothing when the data can pay for everything.

4.4. The multi-method comparison

The same envelope measured for every comparator produces the head-to-head tables that Sections 6 and 12 analyze. As a preview of their shape: on CIFAR-100 under plain augmentation, SimCLR initialization at 2× compute beats the 1.02× prior by +0.85 to +5.0 points between 1% and 10% and converges back to parity by 50% (the full margin envelope is Table 22); SimSiam gains $\leq +0.9$ everywhere at that budget,

Table 6
ViT-tiny: gain of the prior under the plain recipe and under DeiT-strength augmentation, with the amplification ratio, on CIFAR-100 and (last column) Tiny-ImageNet under the plain recipe. Augmentation does not substitute for the prior; it amplifies it and shifts its peak toward more data. Blank: not measured. Three seeds per cell.

	CIFAR-100			Tiny-IN
Fraction	Δ_{plain} ↑	Δ_{DeiT} ↑	Ratio ↑	Δ_{plain} ↑
1%	$+1.4$	$+3.2$	2.4	$+1.4$
2%	$+3.3$	$+6.0$	1.8	$+3.3$
3%	$+6.2$	$+9.5$	1.5	–
5%	$+9.4$	$+16.3$	1.7	$+6.8$
7%	$+11.0$	$+18.7$	1.7	–
10%	$+13.3$	$+21.0$	1.6	$+9.2$
15%	$\mathbf{+14.4}$	$+24.6$	1.7	$\mathbf{+10.7}$
25%	$+13.7$	$\mathbf{+25.1}$	1.8	$+10.5$
100%	$+9.9$	$+13.9$	1.4	$+7.2$

though Section 3.7 shows this to be a property of the budget and not of the method; DINO trails the prior at every fraction below 25%. Raising both comparators to 5× widens SimCLR's lead to +3.1 to +9.2 across the whole envelope, so on convolutional backbones the prior's case is one of cost rather than of accuracy. On attention backbones, and on any backbone under a modern augmentation recipe, the ordering inverts (Section 6).

4.5. Attention: where the prior is dramatic

Table 6 previews the study's most striking axis. ViT-tiny trained from scratch gains +9.9 points from the prior even at *full* CIFAR-100, a scale at which the ResNet backbones are neutral, and under the standard DeiT recipe the gains grow to +25 points, peaking later in the data axis. The best ViT number in the study, 75.3% at full CIFAR-100, is DeiT augmentation *plus* the prior, against 61.4% for the recipe alone. Sections 5 and 7 show this is a feature-side phenomenon that persists to ImageNet scale and grows with model size.

5. The organizing law

A grid of 2,978 cells is only as useful as the account that organizes it, and this section builds that account in four steps. Section 5.1 states the law's form, how the end-to-end gain Δ decomposes into a feature term G and a readout term governed by baseline accuracy, and is explicit about what the law is not. Section 5.2 audits it over the whole grid under a resolvability criterion that keeps uninformative cells from inflating the score. Section 5.3 is the part that distinguishes a law from a curve fit: three episodes in which the account was used to predict unmeasured quantities in advance, including a backbone family it had never seen. Section 5.4 then localizes what each term depends on, which is what makes the fusion outcomes of Section 6 anticipatable rather than merely observed.

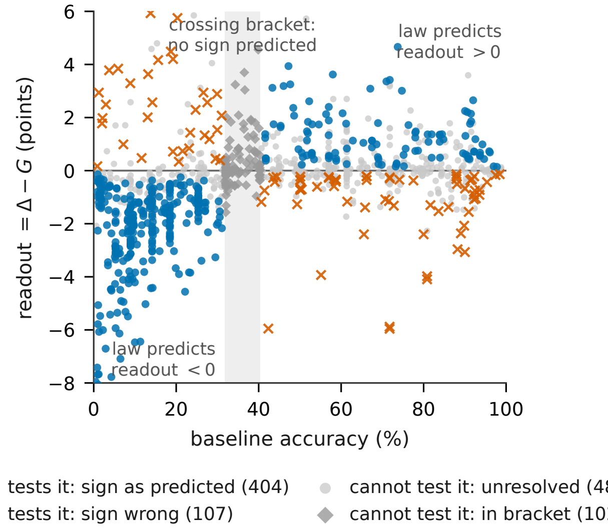

Figure 4: The law, in full: readout $= \Delta - G$ against baseline accuracy for all 1,100 law-scope cells with $\geq$3 seeds on both arms. Shaded band: the measured crossing bracket $[31.8, 40.3]$, inside which no sign is predicted. Of the 511 cells whose readout is resolvable against its own seed-paired uncertainty, 404 fall on the predicted side of zero. The scatter at fixed baseline is substantial ($R^2 = 0.18$, residual SD 2.1 points): the account predicts the sign of this term, not its value.

5.1. Form and origin

For each paired cell we measure the end-to-end gain Δ and the feature gain G (Section 3), and define readout $\equiv \Delta - G$. Across the grid, baseline accuracy is the largest single determinant of this term, and it is the only one whose effect is consistent in sign across datasets and backbones. We are deliberate about how strong that claim is: binned on baseline accuracy alone the readout term has $R^2 = 0.18$, dataset identity explains a further 0.10 of the residual, backbone essentially none (0.004), and the residual standard deviation at fixed baseline is 2.2 points. So this is a first-order trend with substantial scatter, not a curve that pins a cell's readout. What is robust is its *sign*, which is what the rest of this section audits. The term is strongly negative at low baselines (features improve more than accuracy can show: a classifier with five labels per class cannot cash in a better representation), crosses zero in a narrow band we bracket at $[31.8, 40.3]$ percentage points of baseline accuracy, and is positive above the crossing, largest just above it (+1.8 at baseline 39) and decaying toward zero with sufficiency (+0.4 by baselines of 70–80). Fig. 4 shows the full picture; we deliberately fit no parametric form and use only the measured curve for prediction.

We emphasize what the law is not. It is not a theorem; it is an empirical regularity with measured scope, and we found it only after two wrong intermediate laws (a "deficit" law and a multiplicative realization×gain form) were falsified by our own cells and retracted.

Table 7
Scope-wide sign-law audit (regenerable by one command from the released artifact). A cell is resolvable when $|\mathrm{readout}| > 2\,\mathrm{SEM}$; cells inside the crossing bracket make no sign prediction. Uncertainty is seed-paired (Section 3.5), not propagated from independent arms. The rows narrow, step by step, from every measured cell down to the resolvable subset that actually tests the account.

Population	Cells
Cells with paired Δ and G (all interventions)	1,660
in scope, ≥ 3 seed-matched arms	1,100
Backbone variants in scope (architecture families)	9 (5)
Dataset identities in scope	21
Inside crossing bracket (no prediction)	102
Unresolved ($|\mathrm{readout}| \leq 2\,\mathrm{SEM}$)	487
Resolvable (these test the account)	**511**
Sign as predicted	**404 (79.1%)**
Wrong side	107

Table 8
Robustness of the audit. The rate is stable against the resolvability threshold, survives clustering on the units that share a baseline arm, and survives re-fitting the crossing bracket without the dataset being audited, which is the out-of-sample check. It is carried by the left flank, where the account makes a strong prediction, not by the right flank, where it predicts ≈ 0 and is correspondingly weak. Intervals are Wilson 95%.

Audit variant	Cells	Correct	Rate	95% CI
$|\mathrm{readout}| > 1\,\mathrm{SEM}$	709	530	74.8%	[71.4, 77.8]
$> 1.5\,\mathrm{SEM}$	597	463	77.6%	[74.0, 80.7]
$> 2\,\mathrm{SEM}$ (declared)	**511**	**404**	**79.1%**	**[75.3, 82.4]**
$> 3\,\mathrm{SEM}$	382	310	81.2%	[76.9, 84.8]
One vote per (dataset, backbone, fraction)	258	200	77.5%	[72.0, 82.2]
Held out: bracket fitted without the audited dataset	272	218	80.1%	[75.0, 84.5]
Below the crossing	334	293	87.7%	[83.8, 90.8]
Above the crossing	177	111	62.7%	[55.4, 69.5]

Its content is asymmetric: below the crossing it makes strong, falsifiable predictions (large negative readout), while far above the crossing it predicts readout ≈ 0, which is true but weakly discriminating: a naive sign-count over high-data cells would produce a coin flip and wrongly suggest failure. The honest audit therefore counts only cells whose readout is resolvable against its own propagated uncertainty.

5.2. The audit

Table 7 summarizes the audit: of 511 resolvable cells spanning nine backbones and twenty-one dataset identities (the 13 image sources of Table 3 plus the relabeled controls and multi-source views, which share their parent's pixels but are separate label spaces and so are audited separately), 404 (79.1%) fall on the predicted side (Wilson 95% CI $[75.3, 82.4]$; one-sided exact binomial against an even coin, $p \approx 10^{-40}$). Below the crossing, where the account makes a strong prediction, it is $293/334 = 87.7\%$; above it, where it predicts only

Table 9
The ten largest-magnitude resolvable exceptions. Five share the "better features, worse accuracy at sufficiency" signature (G clearly positive, Δ negative, top block); two are PathMNIST cells whose linear evaluation is a known compressed measuring stick on that dataset; three are small-magnitude singletons.

Dataset	Backbone	Variant	Frac.	Base	Δ	G	Readout
Food-101	R18	mag3	50%	71.8	-1.30	$+4.66$	-5.96
Food-101	R18	mag6o	50%	71.8	-0.91	$+4.97$	-5.88
Food-101	MNet	aux	50%	55.1	-0.23	$+3.70$	-3.93
PathMNIST	MNet	aux	20%	89.0	-0.48	$+1.88$	-2.36
PathMNIST	R18	mag3	50%	89.8	-1.36	$+0.71$	-2.07
PathMNIST	R18	mag6o	1%	80.8	$+5.80$	$+9.90$	-4.10
PathMNIST	R18	mag3	1%	80.8	$+5.62$	$+9.60$	-3.98
DTD	ViT	deit-aux	50%	14.2	$+17.30$	$+14.73$	$+2.57$
CIFAR-100	R50	L3	2%	13.1	-0.95	-2.96	$+2.01$
CIFAR-100	R18	L4-λ.3	50%	71.8	-0.73	-0.21	-0.52

that the term is small, $111/177 = 62.7\%$. The 107 exceptions are not scattered noise; a recurring group carries one signature, G clearly positive while Δ is negative at $\geq 20\%$ data on Food-101 and PathMNIST, "better features, worse accuracy at sufficiency". Table 9 lists the largest-magnitude exceptions individually, in the main text rather than an appendix, because where an account fails is part of what it claims.

Table 8 reports what that number does and does not survive. It is not an artifact of where the resolvability threshold was set: the rate moves only from 74.8% over 709 cells at one SEM to 81.2% over 382 at three, so a reader who prefers a stricter or looser criterion reaches the same conclusion. It is not an artifact of the dependence between cells, since collapsing to one vote per (dataset, backbone, fraction) group leaves $200/258 = 77.5\%$. And, the check that matters most, it is not an artifact of estimating the crossing bracket on the same cells we then audit: re-fitting the bracket on the other datasets and auditing each dataset out of sample gives $218/272 = 80.1\%$, statistically indistinguishable from the in-sample figure.

That row audits 272 cells rather than all 511, and the arithmetic is worth stating because the shortfall is not a sampling choice. Re-fitting the bracket on the other datasets makes it *much wider* than the pooled $[31.8, 40.3]$, since a single dataset's cells no longer pin either edge: for CIFAR-100 the re-fitted bracket is $[13.7, 86.8]$. Cells inside a bracket make no sign prediction by construction, so a wider bracket swallows them, and 193 of the 511 leave the held-out set that way, 125 of them CIFAR-100's. A further 46 sit in datasets that cannot receive a fold at all, either because the remaining data cannot bracket them or because fewer than five of their cells survive, which is every population outside the eight largest. The three groups exhaust the total $(272 + 193 + 46 = 511)$. Note the direction of the bias this introduces: the held-out row keeps the cells furthest from the crossing, where the account is most confident, so its agreement rate should be read as an upper bound on out-of-sample performance rather than a like-for-like comparison with the in-sample row. The account generalizes across datasets at the rate it fits within them, which is the one thing about it we can state without qualification.

What the number does *not* survive is disaggregation, and we would rather say so than let a reader discover it. The evidence is overwhelmingly left-flank: below the crossing the sign is right 87.7% of the time, above it 62.7%, barely better than a coin once the term it predicts has shrunk to nothing. Two datasets behave badly out of sample, CIFAR-10 at 50% and PathMNIST at 44%; PathMNIST's linear evaluation is a known compressed measuring stick, since its frozen-feature accuracy sits *below* its own end-to-end accuracy, so we treat its G as unreliable rather than the account as refuted there, but we report both failures rather than excluding the datasets. We do not explain this cluster; we name it as the law's known boundary (Section 14), and the released audit script enumerates every exception, of which Table 9 gives the largest.

5.3. Prediction, not description

Three pre-registered episodes distinguish the law from a post-hoc fit.

(i) A new backbone family from baselines alone. Before any Swin-T linear evaluation existed, we read the readout term off the measured curve at Swin's baseline heights and published bands for its feature gain ($G \approx 9.6$, 8.8, 9.8 and 7.0 at 5, 10, 15 and 25% of CIFAR-100; band +7–+11). The measurements landed at 10.5, 9.3, 10.1, 7.6: four of four in band, each within 0.9 of its point call, against two live falsifiers (attention-intrinsic $G \geq 13$; stabilization-only $G \leq 6$), either of which would have killed the account.

(ii) Four unseen domains. The same procedure applied to satellite, texture, food and histopathology cells straddling the crossing produced eight banded predictions; seven landed in band and all eight had the predicted sign, including the two deliberately hard cases where a small Δ could have hidden a large G (it did not: small Δ meant small G).

(iii) ImageNet scale, the subject of Section 7, where the residual $\Delta - G$ was pre-registered to stay within 1.5 points on every backbone.

5.4. What G is a function of

Controlled crossings localize the law's terms. G is a property of the checkpoint and the evaluation's label space, invariant to which coarse partition relabels the data but cut roughly from 4.2 to 2.9 when training supervision moves from 200-way fine to 20-way coarse labels on identical pixels, fine-grained weak supervision is where the prior has the most feature work to do. The readout term follows task performance, not label budget: engineered cells that lower class count without raising baseline accuracy get no readout boost (a

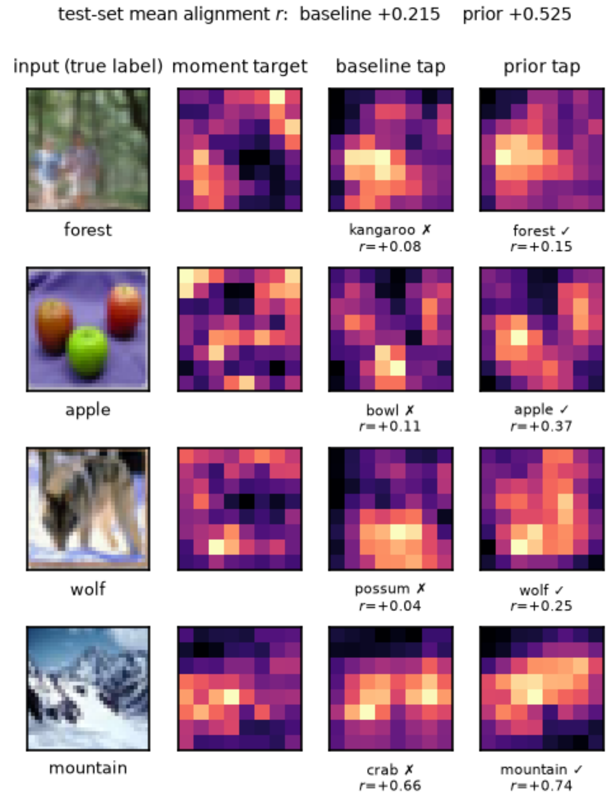

Figure 5: The spectral imprint at the tap, CIFAR-100 at 5%. Columns: input, the fixed moment target, and the tapped-stage energy of the baseline and prior arms. Over the *whole* test set the alignment between tapped features and the target is +0.215 for the baseline and +0.525 for the prior arm; the per-row values are printed beneath each map. The four rows are *selected* cases where the prior arm predicts correctly and the baseline does not, so they illustrate the alignment difference rather than sample it, and only the test-set-wide figure should be read as evidence.

prediction our own account initially got wrong, and the cell that corrected it is in Appendix C). And G depends on the backbone as sharply as on the data: at the same Tiny-ImageNet cell, ResNet-18 carries $G = +2.67$ while MobileNetV3 carries $G = +0.01 \pm 0.23$, a measured zero on the weaker network, so "how much a prior can add" is not a dataset property but a (dataset, architecture) property.

5.5. What the prior leaves in the representation

G is a number, and it is worth seeing what in the representation it corresponds to. The prior's target is a fixed bank (Fig. 2), and its imprint survives to the end of training: at the CIFAR-100 5% cell, where the feature gap is largest on this dataset (+6.3 points), the alignment between tapped features and the bank's oriented-energy target measured over the whole test set is +0.525 for the prior arm against +0.215 for the baseline (Fig. 5). That it survives at all is the part

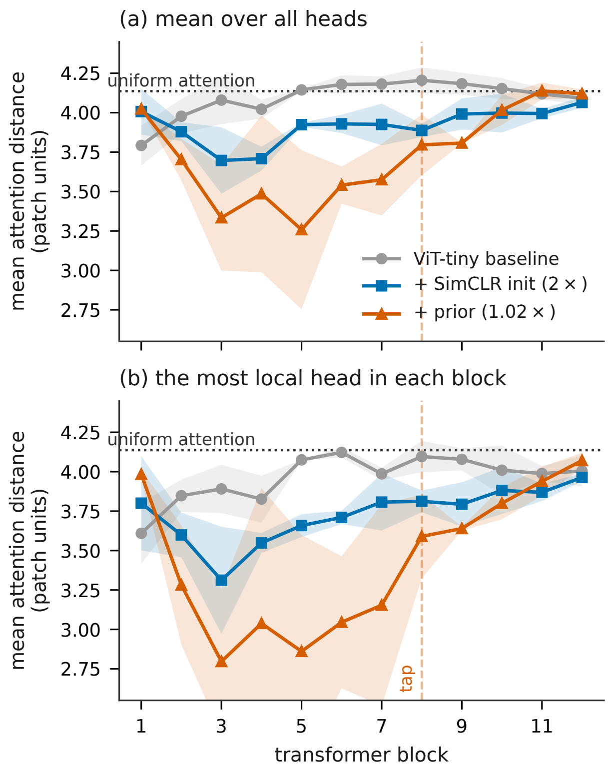

Figure 6: What the prior gives a small transformer: locality. Mean attention distance per block, ViT-tiny on CIFAR-100 at 10%, three seeds (band: ± 1 SD). The dotted line is the distance a *uniform* attention map would give on this 8×8 token grid (4.14 patch units), computed rather than assumed. The baseline sits on that line at every depth: trained from scratch on 5,000 images it never learns to attend locally. The prior pulls the middle blocks well below it, most sharply *upstream* of the tap where its target is imposed, and its most local head reaches 2.80. A SimCLR initialization at $2\times$ the compute produces the same effect, more weakly.

worth noting, because the auxiliary weight decays to exactly zero well before training ends: what the prior changes is where the representation settles, not what the loss is pulling on at the finish.

The same question can be put to the backbone where the prior matters most, and there the answer is unusually legible. A vision transformer has no built-in notion that nearby patches belong together; the standard diagnostic for whether it has learned one is mean attention distance, the average distance between a query patch and the patches it attends to. Fig. 6 measures it for ViT-tiny at CIFAR-100 10%. The baseline's heads sit at 3.79–4.21 patch units across all twelve blocks, straddling the 4.14 that uniform attention would give: on 5,000 images it never discovers locality at any depth. Adding the prior moves the middle blocks to 3.26–3.58 and its most local head to 2.80. The prior does not merely improve the features; it supplies the spatial inductive bias the architecture lacks, which is what a convolutional backbone gets for free and is why the same intervention is worth several times more here.

We had predicted this effect in the *early* blocks, decaying with depth, and that is not what happened: at block 1 the ordering is nominally reversed, the effect peaks at blocks 3–7, and the curves rejoin by block 11. The location is interpretable after the fact, since the target is regressed from block 8 and the locality appears in the blocks that feed it, but we did not predict it and record it as a missed call on a held prediction rather than as a confirmation. A SimCLR initialization produces the same signature at about half the depth-integrated magnitude, which is consistent with the two being substitutes on this backbone.

We are deliberate about how far the qualitative evidence goes. Maps of individual images (Appendix G) show the prior arm concentrating on contours and texture boundaries where the baseline's are diffuse, but those panels are selected on cases the prior gets right and the baseline wrong, so they illustrate rather than demonstrate. A t-SNE of frozen penultimate features shows the same direction and is honest about its size: the silhouette score over all classes moves only from -0.072 to -0.064, so the embedding is *not* visibly better clustered, and a +6.3-point feature gap need not be legible in two projected dimensions. The alignment statistic and the linear evaluation are the measurements; the pictures are illustrations of them.

6. Fusion outcomes: stack, substitute, tax

The grid's central comparative finding is that fusion outcomes are predictable from the *currency* each information source supplies. This section presents the three outcomes in order of how favorable they are, then draws them together. Section 6.1 gives the case where two sources supply different currencies and their gains compound; Section 6.2 the case where they supply the same currency and the second adds nothing, together with the refinement that rescues the rule from its apparent counterexamples; Section 6.3 the one fusion in the grid that is uniformly destructive, and the feature-level measurement that explains why. Section 6.4 collects the three into a single table with the evidence for each row.

6.1. Different currencies stack

DeiT-strength augmentation supplies invariance to nuisance transforms; the prior supplies oriented-energy structure. Fusing both (Table 6) does not trade off: the prior's ViT gain grows 1.4–2.4× under augmentation on CIFAR-100 and 1.4–1.9× on Tiny-ImageNet, and the feature side confirms the mechanism; the prior's G *rises* under augmentation (from 14.9 to 22.2 at 10%), because once nuisance variation is handled the network can commit more capacity to the structure the target teaches. Our own pre-registered bet here was

Table 10
Prior versus SimCLR initialization (2× compute) on ViT-tiny, CIFAR-100, and the effect of recipe. Under plain augmentation SSL catches the prior at ∼5k images; under the DeiT recipe the prior wins at every fraction because the recipe supplies the invariance SSL was selling. Three seeds per cell. A positive entry is a fraction where the prior wins; no better-direction marker is used, since neither sign of a method-versus-method difference is better a priori.

	aux − SimCLR (points)				
Recipe	1%	5%	10%	25%	100%
Plain crop+flip	$+0.7$	$+1.3$	0.0	-1.4	–
DeiT augmentation	$+2.5$	$+7.3$	$+5.2$	$+5.0$	$+2.0$

wrong in the pessimistic direction: we predicted partial substitution and measured strong complementarity.

6.2. Same currency substitutes

SimCLR's objective *is* invariance to an augmentation family. Three measurements pin the consequence. First, prior and SimCLR never stack: on convolutional backbones the combination costs -1.5 points against SimCLR alone, and on ViT it is exactly neutral against the better single their feature gains are the same gain ($G = 14.9$ vs. 13.5 at the same cell), so the second source has nothing left to add. Second, the head-to-head flips with the recipe (Table 10): under plain augmentation SSL overtakes the prior once data is plentiful, but under the DeiT recipe, the recipe anyone training a small ViT would actually use, the prior wins at every CIFAR-100 fraction, by ∼5 points across the mid-band, and the feature side shows why: augmentation lifts the prior's G by $+7.4$ but SimCLR's by only $+3.8$ at 10%. The same flip reproduces on Tiny-ImageNet in the low-data band. Third, SimCLR was not under-equipped: giving its contrastive views DeiT strength makes it *worse* at every fraction (by 4–12 points e2e, halving its G), so its standard configuration was its best shot.

The substitution rule refines rather than universalizes: fusing the prior onto *ineffective* SSL stacks trivially. SimSiam learns almost nothing at study scale ($\Delta \leq +0.9$ everywhere measured), and prior-on-SimSiam recovers the prior's full gain ($+6.2$, $+4.3$ and $+2.0$ over SimSiam alone at CIFAR-100 5, 10 and 25%). The operative rule is therefore *the prior is redundant exactly insofar as the other source already bought the same features*, judged by the other source's own measured gain, not by its family name.

6.3. Fusing into mature features taxes

The one uniformly destructive fusion in the grid is instructive (Table 11). The prior's early shaping, so valuable from scratch, *erases* mature ImageNet features: the tax reaches -17 points where transfer was worth $+13$ to $+18$ on the cells of Table 11, decays to zero at full data (where the initialization no longer carries the performance), and, the decisive measurement, lands almost entirely on G. Where the initialization had little to offer (histopathology), there is nothing to destroy and the tax is a measured zero. Fusion with an information-rich partner must respect what the partner already encodes; a shaping prior does not.

Table 11
The transfer tax and its feature-side origin. Injecting the prior on top of ImageNet weights is destructive in proportion to what the initialization supplied; linear evaluation shows the loss is carried almost entirely by G (features), not by the readout. PathMNIST, where ImageNet features barely transfer, is the null control. Three seeds.

Dataset	Frac.	Transfer adv.	Δ (tax) ↑	G ↑	Readout
CIFAR-10	5%	$+13..+18$	-17.1	-15.8 ± 1.7	-1.3
CIFAR-100	7%	$+10..+18$	-16.4	-16.3 ± 3.1	-0.1
PathMNIST	10%	$+1.8..+2.9$	-0.4	$+0.3 \pm 0.6$	-0.7

Table 12
The fusion taxonomy the grid supports, with the measurement that establishes each row.

Fusion	Currencies	Outcome	Key evidence
Prior + augmentation	structure + invariance	**stack**	Δ amplified 1.4–2.4×; G ↑
Prior + effective SSL	structure ≈ invariance*	**substitute**	combo ≤ best single; equal G
Prior + ineffective SSL	structure + ∼nothing	stack	full gain recovered on SimSiam
Prior + ImageNet init	structure vs. mature features	**tax**	−17 e2e, carried by G; null on shift
Augmentation + SSL	invariance + invariance	substitute	DeiT recipe collapses SimCLR's margin

*Same currency in effect: at matched cells the two sources' feature gains coincide, which is why neither adds to the other.

6.4. The taxonomy

Table 12 states the taxonomy compactly. Its practical force is that the fusion outcome can be anticipated *before* training: measure (or look up) what each source contributes at the feature level, and same-currency pairs will not add.

6.5. Same currency, same images

Everything above infers currency from the *magnitude* of a source's feature gain. That is indirect: two sources could each add fourteen points while improving entirely different images, and calling them the same currency would then be an artifact of comparing scalars. The claim can be tested directly, and we recorded the prediction before running it.

Take three interventions on one backbone, one dataset and one baseline (ViT-tiny, CIFAR-100 at 10%): the prior, a SimCLR initialization, and DeiT-strength augmentation. The taxonomy calls the first two the same currency (they tie end to end, their feature gains are close, and their combination is neutral) and calls the prior and augmentation different currencies (they stack). If that is right, the first pair

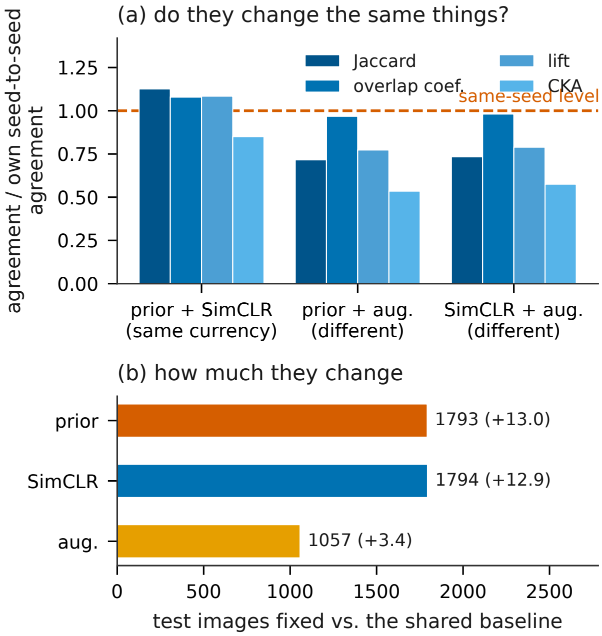

Figure 7: Currency tested directly rather than inferred from magnitudes. All arms are ViT-tiny on CIFAR-100 at 10% with one shared baseline, so the only thing that varies is the fused source. **(a)** agreement between two interventions in *which* test images they fix and *which* representations they learn, each normalized by that intervention's own seed-to-seed agreement, so 1.0 means the two sources differ no more than two seeds of one of them. Four measures are shown because they do not agree on the size of the effect. **(b)** the number of images each intervention fixes: the two same-currency sources fix all but one the same number.

should fix the same test images and learn more similar representations than the second. The confound is that easy images are fixed by everything, so we normalize every cross-intervention agreement by that intervention's own agreement across seeds, which is the most same-currency comparison available.

The prediction held on all four measures (Fig. 7). The prior and SimCLR fix 1793 and 1794 test images respectively, and their normalized agreement on *which* images is 1.13 against 0.72 for the prior with augmentation; centred kernel alignment between their representations gives 0.85 against 0.54. A value above one is worth stating plainly: for the question of which images get fixed, it matters less whether you used a fixed spectral target or contrastive pre-training than which seed you used. That is what a substitute is.

Two honest qualifications. Augmentation fixes far fewer images (1057), and the Jaccard measure penalizes a small set for being small; on the size-robust overlap coefficient the separation shrinks to 1.08 against 0.97, so part of the headline gap is a magnitude effect and only the representation-level measure separates the pairs decisively. And this is one cell on one backbone; it corroborates the taxonomy where the taxonomy

Table 13
The pre-registered scale block. Every falsifier was declared before the first run and none fired, with two qualifications we keep in view: ViT-S's G landed 0.05 below its registered band, and ViT-B's $|\Delta - G|$ of 3.12 exceeds the 1.5 registered for the residual, though on a cell whose evaluation-ceiling caveat was also registered in advance and whose residual is not resolvable against its own uncertainty. G at ImageNet64 uses the fixed-budget evaluation protocol (250 per class shown); ImageNet-100 uses the standard full-train evaluation. Three seeds per cell.

Cell	Registered band	$\Delta\uparrow$	$G\uparrow$	$\lvert\Delta-G\rvert\downarrow$	Verdict
ImageNet64: 1.28M images, 1000 classes					
ResNet-18	Δ: 0.0 ± 0.5	$+0.04$	$+0.05$	0.01	neutral, exactly
MobileNetV3	(none: in bracket)	$+1.95$	$+1.81$	0.14	consistent
ViT-tiny	$\Delta \geq +3$	$+3.23$	$+2.84$	0.39	deficit persists
Swin-T	–	baseline collapses at chance			stabilized by prior
ImageNet-100 @ 224 px, DeiT recipe on ViTs					
ResNet-50	Δ: 0.0 ± 1.0	-0.02	$+0.19$	0.21	neutral at 224 px
ViT-S/16	Δ: $+2..+8$; G: $12.0..12.6$	$+13.00$	$+11.95$	1.05	above band
ViT-B/16	$\Delta \geq \Delta_{\text{ViT-S}}$	$+26.01$	$+22.89^{*}$	3.12^{*}	grows with scale

*The ViT-B aux arm evaluates at its own ceiling (69.8 vs. e2e 69.3), a pre-registered caveat that compresses G; the residual is not resolvable against its uncertainty (±2.4).

already had its cleanest end-to-end evidence, rather than testing it somewhere new.

Takeaway. What a source is worth depends on which currency is scarce, not on how modern the source is: the same free prior is the most valuable intervention we measure on an augmented ViT and the most damaging one on an ImageNet-initialized ResNet.

7. Validation at scale

Everything above could, in principle, have been a small-image regularity. We therefore pre-registered a falsifier block and bought the two scale axes separately: ImageNet64 (1.28M images, 1000 classes, 64 px, 13× the images and 5× the label space of anything else in the grid) and ImageNet-100 at native 224 px with a model-scale curve (ViT-S/16, ViT-B/16, ResNet-50, the ViTs under the standard DeiT recipe). Deviations (reduced epochs; native-resolution transforms) apply equally to both arms of every pair.

Table 13 reports the block. Three conclusions.

Convolutional redundancy at sufficiency is exact and scale-stable. ResNet-18 at 1.28M images gains $+0.04 \pm 0.07$ end-to-end and $+0.05 \pm 0.10$ feature-side; the neutral regime measured to a hundredth of a point at the largest data scale in the study. ResNet-50 at native 224 px repeats it: on three seeds matched across both arms the seed-paired difference is -0.02 ± 0.35, so the pre-registered band (0.0 ± 1.0) holds and its falsifier, which would have fired at $\Delta \geq +1.5$ and made convolutional redundancy a low-resolution artifact, is dead.

The attention deficit is real, persistent, and grows with model scale. ViT-tiny still gains +3.2 points at 1.28M images. At 224 px under the standard recipe, ViT-S/16 gains +13.0, more than ViT-tiny ever gained on CIFAR, and ViT-B/16 gains +26.0, twice ViT-S, with the baseline's seed variance three times the prior arm's. The pre-registered fork read: if the deficit is data-hunger, the bigger model should show the larger gain. It does, by 13 points. Notably, with the prior the *smaller* ViT still outperforms the bigger one at this data scale (78.4 vs. 69.3): right-sizing plus structure beats scale.

The law's predictive form survives. Five of six pairs tie Δ to G within 1.1 points; the sixth (ViT-B) carries a pre-registered evaluation-ceiling caveat and an unresolvable residual. The ViT-S band, derived from the law before the evaluation ran, was hit on its edge (11.95 vs. 12.0–12.6). As an additional robustness check, the ViT-S and ResNet-50 evaluations were independently re-run on a second machine with a from-scratch environment; their means agree with the primary measurement to 0.01–0.05 points.

Takeaway. Scale does not retire the prior: convolutional redundancy at sufficiency survives 1.28M images, while attention's deficit grows with model size rather than shrinking.

8. Buying the comparator more compute

Cost normalization is what makes the comparisons in Section 5 interpretable, and Section 3.7 states the price it charges: a comparator held to its published budget inside our recipe may be undertrained rather than weak, and the two are not separable by construction. That reservation bears directly on our results, so we measured it instead of leaving it as a caveat. Both self-supervised comparators were re-run with four times their pre-training budget, at every fraction from 1 to 25% rather than at one or two, since every other effect in this study moves with the data regime and there is no reason a budget effect should not. Table 14 reports the result, and Sections 8.1–8.5 work through it. It confirms the objection outright, costs us an accuracy claim on convolutional backbones, attaches a budget and a data regime to the claim that survives on attention, and replaces one account of the difference between our two populations with another.

8.1. The near-null was a budget artifact

SimSiam at its published 200-epoch budget gains nothing anywhere on the envelope, −0.11 to +0.87. At 800 epochs it gains up to +4.33. We therefore withdraw the statement that negative-free self-supervision learns almost nothing at this scale: it is budget-starved at the cost we normalized to, and its published default is a poor guide to what it can do on a few thousand images. Its budget response is itself data-dependent, buying most in the mid band (+3.7 at 10%) and nothing at 1%.

Table 14
Buying the comparators more compute. CIFAR-100, ResNet-18, gain over the baseline; three seeds per pre-trained cell, ten for the baseline and the prior at 1, 5 and 10%. Sorted by training cost. The last two rows give the prior minus each comparator at 5× budget, where positive means the free prior still wins.

Intervention	Cost ↓	1%	2%	3%	5%
Prior (this work)	**1.02×**	+1.40	+2.50	+3.68	+5.15
SimSiam, 200 ep	2.00×	−0.04	−0.11	+0.11	+0.13
SimCLR, 200 ep	2.00×	+2.25	+4.88	+5.99	+9.05
SimSiam, 800 ep	5.00×	−0.11	+0.19	+0.54	+1.90
SimCLR, 800 ep	5.00×	**+4.52**	**+8.33**	**+10.82**	**+14.38**
Prior − SimSiam-800		+1.51	+2.31	+3.14	+3.25
Prior − SimCLR-800		−3.12	−5.83	−7.14	−9.24

Intervention	Cost ↓	7%	10%	15%	25%
Prior (this work)	**1.02×**	+4.87	+3.75	+2.55	+0.16
SimSiam, 200 ep	2.00×	+0.87	+0.61	+0.17	−0.03
SimCLR, 200 ep	2.00×	+9.86	+8.76	+6.64	+2.81
SimSiam, 800 ep	5.00×	+3.39	**+4.33**	+3.65	+0.79
SimCLR, 800 ep	5.00×	**+13.88**	**+10.92**	**+8.36**	**+2.77**
Prior − SimSiam-800		+1.48	−0.58	−1.10	−0.63
Prior − SimCLR-800		−9.01	−7.17	−5.82	−2.61

That data-dependence is why the envelope was necessary, and the point is methodological rather than incidental. Had we run only 5 and 10%, the two fractions where the objection bites hardest, we would have reported that the prior beats 5× SimSiam at the scarcer fraction and ties at the other. The full envelope shows a crossover instead: the prior wins from 1 to 7% by up to 3.25 (2.6 to 12.2 standard errors) and *loses* from 10 to 25%, most clearly at 15% (−1.10, 5.2 standard errors). A two-point read would have missed a significant reversal, which is the same sparse-grid error we identify in our own earlier readings elsewhere in this paper.

8.2. On convolutions the prior's case becomes a cost case

Against the stronger comparator the prior loses at every fraction, including the scarcest. SimCLR at 5× beats it by 3.12 to 9.24 points, at 8 to 37 standard errors, from 500 images to 12,500. Our earlier positioning noted near-parity with self-supervision at 1%; that was a statement at 2× and it does not survive 5×. On convolutional backbones the prior's case is therefore a cost case and not an accuracy case, and we say so without qualification.

8.3. Attention, given the same test

Leaving the attention comparison at 2× while raising the convolutional one to 5× would reproduce

Table 15

The prior minus a 5×-budget SimCLR initialization, on both populations, under DeiT augmentation on ViT-tiny. Positive means the free prior wins. The ordering tracks the data fraction, not the absolute image count: 5,000 images is 10% of CIFAR-100 but 5% of Tiny-ImageNet, and the two disagree there while agreeing at every matched fraction.

Prior − SimCLR-800	5%	10%	25%
CIFAR-100	−1.57	+0.71	+1.47
Tiny-ImageNet	−1.35	−0.31	+1.71
verdict	comparator	level	prior

exactly the asymmetry this section exists to remove, so we ran it there too. Against ViT-tiny under the DeiT recipe, 5× SimCLR gives 30.45, 40.57 and 55.85 at 5, 10 and 25%, against the prior's 28.87, 41.29 and 57.32. The comparator *wins* at 5% by 1.57 (3.9 standard errors, six seeds), the two are level at 10% (+0.71, 1.2), and the prior wins at 25% by 1.47 (7.2). The prior is not overturned on attention as it is on convolutions, but the honest form of the claim names its budget: at 2× the prior leads at every fraction, and at 5× only in the higher-data band. Section 8.4 shows the same boundary on a second population.

The shape of that result contradicted a prediction we had recorded before running it, and the error is diagnostic rather than embarrassing. We expected the prior to hold at the scarce fractions and 25% to be the cell most likely to flip; the reverse happened. What the extra budget buys makes the reason visible: +8.32, +4.51 and +3.53 at 5, 10 and 25%, decreasing with data, because 200 epochs over 2,500 images is fewer than four thousand steps. We had assumed the comparator was data-starved at 5% beyond what compute could fix; it was step-starved. On convolutional backbones the same increment peaks in the mid band instead, so even the budget response is backbone-dependent.

8.4. A second population, and the account that failed

Repeating the comparison on Tiny-ImageNet gives the same ordering at every matched fraction (Table 15). The comparator wins at 5% on both (3.9 and 3.7 standard errors), the two are level at 10% on both, and the prior wins at 25% on both (7.2 and 5.1). So at five times the comparator's pre-training budget the prior leads in the higher-data band and loses in the scarcer one, and that pattern is not specific to a dataset.

Getting there cost us a prediction and an account, and both are worth reporting. Having measured only 5 and 10% on Tiny-ImageNet, where the comparator was ahead and level, we wrote that the prior led nowhere on that population and that the two datasets told different stories. We then predicted the 25% cell at 40–44, explicitly against our own method. It landed at 36.53: the prior wins by 1.71, our band missed low by 3.5, and the falsifier we had recorded for that cell fired. The tin envelope has a crossover after all, and the "different stories" reading was an artifact of a two-point grid, which is the same error this section has already documented once.

The account we had proposed also fails, and its replacement is what the matched-fraction table shows. We had expected the two populations to line up by *image count*. They do not: at a matched 5,000 images CIFAR-100 is level while Tiny-ImageNet has the comparator ahead. They line up by *fraction* instead. Five thousand images is 10% of CIFAR-100 but 5% of Tiny-ImageNet, so matching on image count compares different points on two different learning curves. What determines which source wins is where a cell sits on its own dataset's curve, not how many images it holds.

One measurement in this pass we report without explaining. At Tiny-ImageNet 25%, four times the pre-training made the comparator *worse*: 38.48 at 2× against 36.53 at 5×, a drop of 1.95 ± 0.42. It is the only cell in the study where extra pre-training budget costs accuracy, and it echoes our earlier finding that DeiT-strength contrastive views also hurt. We note it rather than fit an explanation to one cell.

8.5. What the budget actually buys

The comparison so far is end-to-end, while the taxonomy is a claim about *what* a source supplies rather than how much accuracy it yields, so we probed these cells under the frozen-feature protocol used throughout.

On CIFAR-100 the feature gain of 5× SimCLR is +22.24, +21.27 and +20.77 at 5, 10 and 25%, against the prior's +21.37, +22.19 and +23.05. The two orderings agree cell by cell: the comparator's feature gain is larger where it wins end-to-end, level where the two are level, and smaller where the prior wins. The prior's feature gain *rises* with data while the comparator's *falls*, and they cross between 5 and 10%, which is where the accuracy ordering crosses. Extra pre-training budget therefore buys more of the same currency rather than a different one, and the flip is a property of the representations rather than of the classifier reading them. This is the substitution outcome measured on a budget axis, the one axis the taxonomy had not been tested on.

Including the second population makes the agreement stronger and supplies the mechanism the image-count account failed to give. Across both datasets and all six measured fractions, whichever intervention has the larger frozen-feature gain is the one that wins end-to-end: the sign of $G_{\mathrm{SSL}} - G_{\mathrm{prior}}$ matches the sign of the accuracy difference at six cells out of six. The sharpest of these is the cell that overturned our own prediction. At Tiny-ImageNet 25%, where the prior wins by 1.71, its feature gain is +14.27 against the comparator's +12.43; the difference of 1.84 and the accuracy difference of 1.71 agree to within 0.13, so that outcome is feature-side almost in its entirety. And the population term is not mysterious once measured. At a matched 5,000 images the two datasets diverge because the *prior's* contribution diverges: its feature gain is +22.19 on CIFAR-100 and +15.67 on Tiny-ImageNet, while the comparator's is +21.27 and +17.72. Moving to the 200-way 64 px task costs contrastive pre-training about 3.5 points of feature gain and costs the fixed spectral target about 6.5. A hand-crafted oriented-energy prior supplies proportionally less of what a finer label space at higher resolution requires, and that, rather than the number of images, decides which source wins.[1]

The second comparator extends the same agreement. Probing SimSiam at 5× across all eight fractions gives a feature gain that sits below the prior's from 1 to 7% (−5.3 to −1.9 points of difference), crosses at 10%, and rises above it at 15%, matching the end-to-end ordering reported in Section 8.1 at seven of eight cells; the two that are ambiguous are the ones where both quantities are near zero. Across two self-supervised methods and two backbone families, the source with the larger frozen-feature gain is the one that wins.

One result in this pass we report without an account. Probing the convolutional budget envelope shows the prior's own residual tracking the fitted branch closely (−2.76, −2.65, −2.07, −1.11 below the crossing, then +0.06, +0.21, −0.18, −0.28 above it), while 5× SimCLR departs from it, running positive from 3 to 15% (+1.32, +3.11, +3.58, +2.23, +1.81) at baselines where the branch is negative or near zero. Two things narrow what the anomaly can be. It is not a property of self-supervised initialization as such: 5× SimSiam, initialized the same way and evaluated identically, stays within $[-0.42, +0.86]$ at every fraction. And across all sixteen self-supervised convolutional cells the residual correlates with the size of the intervention itself ($r = +0.52$ between Δ and the residual), so what departs from the branch is large gains rather than a particular method. We recorded in advance that these initializations lie outside the scope in which the branch was estimated, so we do not count the departure against it, and we prefer to leave it as a measured regularity than to fit an explanation to sixteen points.

Takeaway. A comparator's budget is part of the claim. Two of our own accuracy claims did not survive this section and are now cost claims; the finding that did survive is that heavy augmentation, not the prior, is what collapses contrastive pre-training's advantage.

[1] The Tiny-ImageNet comparator probes were computed on a different machine from the cells they pair with. Cross-machine agreement under this protocol was verified elsewhere in the study to within 0.05 points, but we note the mixed provenance rather than leave it implicit.

Table 16
Does a second source pay? Fusion gain is the fused arm minus the better single source, at matched data. Sorted by data fraction. Three seeds per cell, ResNet-18; the champion configuration is used verbatim throughout. Splitting one instrument's bands pays at every fraction; pairing two satellites does not pay at any.

	EuroSAT-MS (one instrument)		So2Sat (two satellites)	
Data	$\lvert\Delta_{acc}\rvert \downarrow$	Fusion $\uparrow$	$\lvert\Delta_{acc}\rvert \downarrow$	Fusion $\uparrow$
1%	6.68	+1.40	26.31	−0.57
2%	1.07	+2.96	33.77	−0.14
5%	0.58	+1.34	30.61	−0.73
10%	0.66	+1.15	28.51	−0.11
25%	0.11	+0.45	25.98	±0.00

9. Fusing genuinely different sources

Everything to this point fuses one knowledge source into one image stream. This section asks whether the same rule governs fusion in the sense the term usually carries: two *sensors*, and two *classifiers*. Section 9.1 takes visible and non-visible Sentinel-2 bands as separate sources; Section 9.2 fuses trained models at the decision level. Both are tests of transfer rather than restatements: the rule was derived from training-time fusion of a prior into a network, and neither of these is that.

9.1. Two sensors, and when a second one pays

We build two multi-source populations that differ in how far apart the sources are, because that turns out to be the variable that matters.

EuroSAT-MS splits *one* instrument: Sentinel-2's 13 bands become **visible** (3), **non-visible** (the other 10, spanning red-edge, near-infrared, short-wave infrared and cirrus) and **sensor-fused** (all 13). **So2Sat LCZ42** pairs *two satellites*: Sentinel-1 synthetic-aperture radar (8 channels) with Sentinel-2 optical (10 channels), over 17 local-climate-zone classes. The second is the harder and more honest test, since radar backscatter and optical reflectance share no detector and no physics. In both, tiles and fixed subset indices are identical across the sources and the configuration is the champion verbatim with only the source changed, so nothing but the input varies.

One instrument's bands are complementary; two satellites are not. Fusing EuroSAT's band sets beats the better single source at every fraction, by +0.45 to +2.96. Fusing So2Sat's two satellites never does: the 18-channel arm lands between −0.73 and ±0.00, at or below optical alone throughout. Radar is not uninformative in isolation, reaching 35–53% against a 5.9% chance rate, but adding it to optical buys nothing here. Had we run only the multispectral population we would have concluded that a second sensor stacks, and reported a result that the first genuinely cross-modality test contradicts.

What separates the two populations, and what we cannot yet separate. The arms differ enormously in how well matched they are: the accuracy asymmetry between the two sources is 0.11–6.68 points on EuroSAT-MS and 25.98–33.77 on So2Sat, and the population with the matched arms is the one where fusion pays. That is consistent with Equation 9, our decision-level fit on CIFAR-100 classifier ensembles, which is dominated not by diversity but by exactly this asymmetry term.

We had reported a correlation of $r = -0.80$ between asymmetry and fusion gain across the ten cells of Table 16, and we withdraw it. The arithmetic is right but the statistic is not: within EuroSAT-MS the correlation is $+0.08$ and within So2Sat it is -0.05, so it is carried entirely by the gap between two clusters that do not overlap on either axis. Its effective sample size is two populations, not ten points. And those two populations differ in *two* ways at once, since the low-asymmetry population splits one instrument's bands while the high-asymmetry population pairs two satellites with no shared detector or physics. Nothing in this design separates "the sources are too unequal" from "the sources are cross-modal", and we should not have offered a correlation coefficient as though it did.

What the experiment supports is therefore a contrast, not a predictor: fusion paid at every data scale within one instrument and never paid across two satellites, and the two settings differ in both source asymmetry and modality. Distinguishing the two would need a third population that breaks the confound, such as two well-matched cross-modal sources or two badly matched same-instrument ones, and we have not run it.

The prior itself transplants to both sensors, including radar. On optical it produces the study's usual envelope, $+1.13$, $+1.91$, $+1.19$, $+0.68$ and -0.56 at 1, 2, 5, 10 and 25%: positive while data is scarce, decaying to neutral at sufficiency. On SAR it gives $+1.24$ at 1% and $+2.59$ at 5%, its largest So2Sat gain, then -0.46 and -0.24 at 10 and 25%. We had registered in advance that radar was where a fixed oriented-energy target might fail, being speckle-dominated and carrying no photographic edge statistics, with a falsifier at $\Delta \leq -1.0$ on two fractions. It did not fire. Only the 2% SAR cell falls below that, and both of its arms are unstable (seed spreads of 3.5 and 5.5 points against 0.1–1.1 everywhere else in the column), so we read it as variance rather than as effect and flag it as such. On the multispectral population the picture is the same at 5%, where the prior adds $+1.63$, $+2.60$ and $+3.30$ on the visible, non-visible and fused sources.

Whether the prior amplifies a second sensor is backbone-dependent, and we claim only what both backbones support. On EuroSAT the ResNet gain rises with sensor count while on ViT-tiny it inverts, $+5.14$, $+2.69$, $+1.75$, tracking baseline weakness (84.80, 90.19, 91.52) as the rest of this study does. The defensible statement is that a second source does not absorb the prior on either backbone. On the arms that carry the prior, EuroSAT's fusion gain is $+1.92$, $+1.46$, $+2.04$, $+1.01$ and $+0.46$ against the baseline arms' $+1.40$, $+2.96$, $+1.34$, $+1.15$ and $+0.45$: the prior and the second sensor coexist rather than substituting.

The feature side carries part of it. Under identical frozen-feature evaluation the prior's G on So2Sat at 5% is $+1.31$, $+1.54$ and $+0.88$ on radar, optical and fused, against end-to-end gains of $+2.59$, $+1.19$ and $+1.35$; on EuroSAT at 5% G is $+0.98$, $+1.29$ and $+1.16$ against $+1.63$, $+2.60$ and $+3.30$. So the gain is genuinely feature-level in part, and while data is scarce the readout term supplies the rest. Every baseline in these two populations sits between 43.5 and 98.3, far above the crossing, which is exactly the regime where Section 5.2 reports the readout term decayed to nothing and its sign consequently uninformative; the residuals here run from -0.96 to $+1.28$ and we do not read signs from them.

9.2. Two classifiers: decision-level fusion

Multi-classifier fusion is a different level again, and the rule was not derived there. From the CIFAR-100 5% checkpoints, with no further training, we fuse every pair of arms (i, j) by averaging their class posteriors, $p_{ij}(y \mid x) = \frac{1}{2}\big(p_i(y \mid x) + p_j(y \mid x)\big)$, and characterize each pair by three scalars on the test set $\mathcal{T}$: the *gain* over the better single arm,

$$g_{ij} \;=\; \mathrm{acc}(p_{ij}) \;-\; \max\!\big(\mathrm{acc}(p_i),\, \mathrm{acc}(p_j)\big), \tag{7}$$

the *disagreement* rate, the classical diversity measure [49],

$$d_{ij} \;=\; \frac{1}{|\mathcal{T}|} \sum_{x \in \mathcal{T}} \mathbf{1}\big[\, \hat{y}_i(x) \neq \hat{y}_j(x) \,\big], \tag{8}$$

where $\hat{y}_i(x) = \arg\max_y p_i(y \mid x)$, and the *accuracy asymmetry* $a_{ij} = |\mathrm{acc}(p_i) - \mathrm{acc}(p_j)|$. Here g_{ij} and a_{ij} are accuracy points and d_{ij} is a percentage of test images, so β_d below converts one into the other. The self-supervised arm here is a 50-epoch pre-training checkpoint rather than the 200- or 800-epoch ones of Section 8, because this analysis needs two arms of *matched accuracy* so that the asymmetry term does not dominate, and that is the budget at which SimCLR lands within a point of the prior. The budget lesson of Section 8 applies here too: a longer-pre-trained arm would ensemble differently, and we have not measured that.

The first thing the data say is a caution rather than a confirmation. Least squares over the $n = 28$ pairs gives

$$\hat{g}_{ij} \;=\; \beta_0 \;+\; \beta_d\, d_{ij} \;+\; \beta_a\, a_{ij}, \tag{9}$$

with $\beta_0 = -2.74 \pm 0.48$, $\beta_d = +0.089 \pm 0.011$ and $\beta_a = -0.341 \pm 0.047$ ($R^2 = 0.74$, residual standard

Table 17
Decision-level fusion on CIFAR-100 at 5%, restricted to pairs whose two arms are within one accuracy point of each other so that the information source, rather than an accuracy gap, is what differs. Gain is over the better single arm; disagreement is the fraction of test images on which the two arms predict different classes. Averaging posteriors requires no further training.

Pair	Gain ↑	Disagreement
Prior + SimCLR (50 ep)	$+1.20$	47.2%
SimCLR (50 ep) + prior, wide bank	$+1.13$	47.2%
Prior + prior, wide bank	-0.04	28.1%

deviation 0.42 points, every coefficient resolved at $|t| > 5$). Both terms matter, but the dominant one is not diversity: a one-point accuracy gap costs $|\beta_a| = 0.34$ points of gain, which four points of extra disagreement are needed to recover. Pairing a strong arm with a weak one destroys the gain regardless of what either contributes. Restricting to pairs within one point of each other, where currency is actually the free variable, leaves the three pairs of Table 17.

Two models built on the same knowledge source disagree on 28% of test images and ensembling them buys nothing; pair either with a self-supervised model and disagreement rises to 47% and ensembling buys ∼1.2 points. That is the currency rule with its mechanism visible, and it is what the classical account would predict.

We report the qualification with equal weight, because it bounds the taxonomy rather than extending it. The prior and SimCLR do *not* stack at training time (Section 6.2), yet they ensemble usefully at decision time (+1.20). Redundancy in the sense of "jointly training on both adds nothing" therefore does not imply redundancy in the sense of "their errors are correlated". The substitution result is a statement about training-time fusion, and we do not claim it transfers to decision fusion. With three matched pairs this is suggestive rather than established, and we say so.

9.3. A procedure for deciding whether to fuse

The practical content of the taxonomy is that the outcome of a combination can be estimated *before* the combined system is built, from measurements of each source alone. We state that as a procedure rather than as a finding, since it is what a practitioner would actually run.

Table 18 states it. Three properties are worth drawing out. It is *cheap*: the only cost beyond training each source once is two linear evaluations on frozen features, which run in minutes on cells that took hours to train. It is *predictive rather than retrospective*, which places it alongside transferability estimation [28], with

Table 18
Deciding whether to fuse two sources, before training the combination. Every quantity is measured on the sources separately; the only cost beyond training each source once is two frozen-feature evaluations.

Inputs: sources A, B; a baseline arm; a labeled split	
1	Train the baseline, A alone and B alone. Record each arm's end-to-end gain and declared compute multiple.
2	Evaluate all three frozen representations with an identical linear protocol. This gives G_A and G_B, each source's feature-level contribution.
3	Compare the two arms' accuracies. If one is far weaker, treat the combination as unpromising: this term dominates our decision-level fit (Eq. 9), and the sensor population whose arms are matched is the one where fusion paid. Weakest-supported step, since our two sensor populations differ in modality as well as in asymmetry (Section 9.1); use it to rank candidates, not to reject one.
4	If either source's gain is ≈0, it supplies nothing to overlap: the other source's gain is available in full, and the combination is not the question.
5	If $G_A \approx G_B$ *and* both arms improve the same cells, the sources carry the same currency. Expect **substitution**: the combination will not exceed the better single arm. Take the cheaper source.
6	If G_A and G_B differ in kind, expect **stacking**, and a combined gain larger than either alone.
7	If one source is a mature initialization, do not inject a shaping prior on top of it. Expect a **tax** proportional to what the initialization was contributing, measurable in advance as that source's advantage over training from scratch.

the difference that those methods score a single pre-trained source for a target task while this scores a *pair* for combination. And it is *falsifiable*: step 5 asserts that equal feature gains imply no benefit from combining, which Sections 6.2 and 9.2 both test and which the training-time-versus-decision-time discrepancy above already qualifies.

The honest limit is step 5's antecedent. We compare feature gains by magnitude and by which cells improve, not by a formal decomposition of the information each source carries. Partial information decomposition [26, 27] is the principled version of that comparison, and an estimator that scaled to these representations would replace our proxy with a quantity that means something on its own.

Takeaway. The currency account extends past training-time fusion: across sensors and across classifiers, how asymmetric two sources are predicts whether combining them pays, and a second source that is merely weaker rather than different does not.

10. Does any of this hold off classification?

Every number to this point is top-1 accuracy. That is a real limit on what the benchmark can claim, because the auxiliary target is itself a *dense* map — oriented energy at every location — and a reader is entitled to ask whether the fusion outcomes and the law $\Delta = G +$ readout are properties of the method or of the metric. This section answers that on semantic segmentation.

State the bias before the result. A dense spatial target on a dense spatial task is the most favourable venue this prior could be given. A *positive* result here would therefore be weak evidence of generality. What makes the experiment worth running is that its headline finding is negative, and negatives are not flattered by a favourable venue.

Setup. Six populations spanning label-space width and domain: **PASCAL VOC-2012** (21 classes, augmented split), **Cityscapes** (19, driving, half resolution), **FoodSeg103** (104, texture-dominated), **ADE20K** (150, scene-centric, no background class), **Pascal-Context** (254), and a **Swin-Tiny** arm on VOC. The encoder is the study's ResNet-18 at output stride 8 with an FCN head, so the prior's bank, tap, λ schedule and head normalization transfer with no re-design and no new hyper-parameter. Figure 8 summarises the result. 6 populations $\times$ six fractions $\times$ two arms $\times$ three seeds = 216 cells, all completed.

Pascal-Context is the control that carries most of the weight: it re-annotates the *identical* VOC images at 254 classes instead of 21, so it separates a label-space effect from a pixel effect exactly as CIFAR-100-super does on the classification side.

Two measurement decisions, both of which we got wrong first. Absolute mIoU is not comparable across these populations: their baselines differ eightfold (51.5 on VOC at full data, 6.4 on Pascal-Context). Comparing raw Δ across them made the label-space effect look like a 13$\times$ collapse when in relative terms it is at most $\sim$1.6$\times$. We report both, and read comparisons off the relative column.

Second, and more consequential: *readout must be evaluated on the head's own classification scale, not on mIoU.* The crossing bracket [31.8, 40.3] was measured in accuracy. mIoU is a far harsher statistic — it averages per-class IoU with equal weight, so classes the model never learns dominate it — and a cell at 4.7 mIoU may be classifying 72% of its pixels correctly. Scoring the law on mIoU places every dense cell deep on the left flank and predicts a negative readout everywhere; scoring it on pixel accuracy places them on the decayed positive branch, which is where they are. We registered this caveat in advance and then violated it in our own prediction, which is how it was found.

10.1. Target–task alignment is not where the value comes from

This is the section's main result and it is a negative one. VOC at 1% is 107 images over 20 classes, about five per class — the same supervision density at which classification shows a universal $\approx +1.5$ floor (CIFAR-100 +1.42, Tiny-ImageNet +1.49). Dense prediction at that density returns +0.39 **mIoU** (6.2%), and the frozen-feature probe agrees ($G = +0.31$), so it is not a readout artifact.

Table 19

The dense envelope. Δ in mIoU points, with Δ as a percentage of that cell's own baseline beneath it. Three seeds per cell, ResNet-18 with an FCN head at output stride 8 (Swin-Tiny in the last block), 200 epochs at every fraction — the same budget the classification recipe uses. Pascal-Context is VOC's own pixels relabelled to 254 classes.

Population	1%	2%	5%	10%	25%	100%
VOC (21)	+0.39	+0.97	+1.15	+1.61	+2.34	+0.46
rel.	6%	12%	9%	9%	8%	1%
Cityscapes (19)	+0.49	+0.41	+0.20	+0.68	+0.73	+0.48
FoodSeg103 (104)	+0.12	+0.20	+0.26	+0.43	+0.56	−0.12
ADE20K (150)	+0.18	+0.20	+0.38	+0.38	+0.37	−0.29
Pascal-Ctx (254)	+0.11	+0.13	+0.16	+0.10	+0.16	+0.17
rel.	11%	10%	9%	4%	5%	3%
Swin-Tiny, VOC	+0.42	+1.29	+1.76	+0.90	+0.75	+1.31
rel.	9%	21%	22%	8%	5%	5%

So the venue built to flatter the prior — a dense target supervising a dense task — pays *less*, not more. Whatever the prior supplies, it is not "structure that happens to match the output space." That makes the mechanism claim in Section 6 cleaner rather than weaker: the prior injects oriented-energy structure into the representation, and what the head does with it afterwards is a separate question.

10.2. The regimes transfer; the magnitudes are metric-bound

In absolute mIoU the dense gains look tiny beside the classification ones. In relative terms they run 1–22% of baseline across the 1–25% band, which is the same range classification occupies (CIFAR-100 at 5% is +5.15 on 25.36, i.e. 20%). The shape transfers too: the envelope is unimodal, peaking at 25% on VOC (+2.34) and FoodSeg103 and at 10% on ADE20K, and it decays to neutrality or below at full data on three of six populations (VOC +0.46, i.e. 0.9%; FoodSeg −0.12; ADE20K −0.29). Structural neutrality at sufficiency, which the $\lambda \to 0$ schedule is supposed to guarantee by construction, therefore survives the change of task and metric.

A validity note that changed a conclusion. We first ran this grid at 50 epochs and reported an envelope that was monotone rising on every population, with no right flank. That was an artifact. At 200 epochs the VOC 10% baseline rises from 7.23 to 18.42 mIoU (+155%), and the 50-epoch curves were still climbing at ~0.7 mIoU/epoch when the cosine schedule annealed the step size away — their apparent convergence was the schedule, not the model. Re-pinning the dense recipe to 200 epochs, matching the classification budget rather than choosing one to suit the task, moved $\Delta(\mathrm{VOC}, 100\%)$ from +2.57 to +0.46 and turned a nominal falsification of structural neutrality into a confirmation of it. We report this because an under-trained instrument does not merely add noise: here it produced the opposite scientific answer, twice.

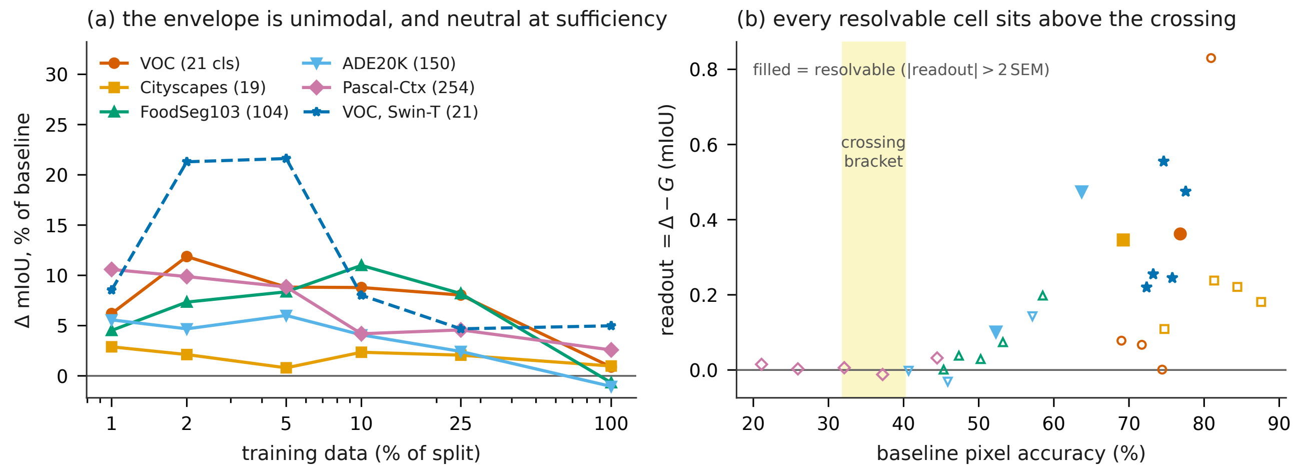

Figure 8: Semantic segmentation, six populations, 200 epochs, three seeds per cell. **(a)** Δ as a percentage of each cell's own baseline, not in absolute mIoU: the populations' baselines differ eightfold, so absolute deltas are not comparable across them. Read this way the envelopes are unimodal and occupy the same 5–20% band classification does, and three populations go neutral or negative at full data. **(b)** The readout term against the head's own classification scale (pixel accuracy), not against mIoU — the crossing bracket is an accuracy bracket, and scoring the law on mIoU places every dense cell on the wrong flank. Filled markers are the cells whose readout is resolvable against its own uncertainty. Every one of them sits above the bracket, which is why segmentation confirms the law's positive branch and cannot test its negative branch.

10.3. The law on a different task and a different metric

Of the 30 cells carrying both Δ and G (full-data cells are excluded: there the probe's labels *are* the cell's labels, so no split is interpretable), 2 sit inside the crossing bracket where the law makes no sign call and 21 have $|\text{readout}| \leq 2\,\text{SEM}$, which on the right flank is expected rather than disappointing — Δ and G are both near zero there and the sign of their difference is noise. That leaves 9 cells that actually test the law, and 9 of 9 fall on the predicted side.

The honest qualification is that all 9 sit *above* the crossing (pixel accuracy 52–78), so segmentation confirms the law's positive branch on a new task and a new metric and tests its negative branch not at all. Pascal-Context was built to supply exactly that test — its trained pixel accuracy is 21.1 and 25.9 at 1–2%, clearly below the bracket, where a negative readout is required. It returns +0.02 and +0.00: zero, not negative. The pre-registered falsifier needed a clearly positive readout and so did not fire, but the prediction was not confirmed either, and we record the cell as *unresolved*. The obstruction is structural: a population that could settle it needs abundant labels, low pixel accuracy, *and* a Δ large enough for readout to resolve, and none of our six has all three.

10.4. Attention, narrowed

On classification the prior is worth roughly twice as much to a hierarchical-attention backbone as to a convolutional one ($G_{\text{swin}} = 7.6$–10.5 against $G_{\text{r18}} = 3.55$–6.26 on CIFAR-100). On dense prediction that advantage survives only in a narrow band. Relative feature gain, Swin against ResNet-18 on identical images: 4.6, 20.7, 17.4, 7.0, 2.0% at 1–25%, against 6.3, 13.4, 10.8, 8.1, 5.8%. Swin leads at 2% and 5% by $\sim$1.6$\times$ and trails at 1%, 10% and 25%; by 25% its G has collapsed to +0.27 against convolution's +1.51. At full data it still gains (+1.31, 5% relative) where the convolutional arm is near-neutral (+0.46, 1%), which does mirror the classification result.

We state this narrowly on purpose. An earlier, under-trained version of this experiment supported the opposite conclusion — that the attention deficit does not transfer to dense prediction at all — and a four-fraction reading of the corrected data supported a broader one. Both were over-readings of the fractions then available. Swin is not ViT, this is one dataset, and these are AdamW diagnostic cells; what they establish is that the attention advantage is task- and regime-dependent, not that it is absent.

10.5. The label-space effect is much weaker than it looks

VOC and Pascal-Context are byte-identical pixels at 21 and 254 classes. In absolute mIoU the gap is large at every fraction, and at full data VOC's +0.46 against Pascal-Context's +0.17 invites a conclusion about label granularity. Normalized, it mostly disappears: 0.9% against 2.6% at full data, 9% against 9% at 5%, and Pascal-Context is *higher* at 1% (11% against 6%). Relative feature gain behaves the same way. The defensible statement is a mid-band tendency for the coarser label space, not the collapse the absolute numbers suggest —

Table 20
Target and mechanism controls on CIFAR-100 with ResNet-18 (each row is the champion configuration with one ingredient varied; Δ vs. the shared baseline, three seeds). These cells predate the champion's final weight schedule and so are not directly comparable to the envelope of Table 4; they are internally comparable to each other, which is what an ablation requires. The gain needs the moment structure specifically: random fixed targets do nothing, a $2\times$-cost learned teacher does nothing anywhere on an eight-point envelope, and the classical HOG descriptor recovers about half.

	Δ at fraction ↑		
Aux target	5%	10%	Note
Moment magnitude (ours)	$+3.31$	$+2.81$	phase-invariant energy
Structure tensor	–	$+1.78$	mild nonlinearity
Steerable harmonics	–	$+1.01$	principled rot.-inv.
Rotation-invariant energy	–	$+0.84$	
Oriented edges (Gabor)	–	-0.24	raw edges hurt
2nd-order invariants	–	-2.97	over-processed
HOG target	$+1.22$	$+1.37$	descriptor helps; moments $2\times$ better
Learned teacher (FitNets)	-0.36	$+0.16$	$2\times$ cost, $\sim$0 on 8-pt envelope
Random fixed maps	$+0.01$	$+0.14$	kills "any aux signal"
Cosine loss (not MSE)	–	$+0.46$	magnitude scale matters

and it is only visible as such because the control holds the pixels fixed.

11. Ablations and controls

The benchmark's defensibility rests on knowing *which* ingredient carries the gain. We ablate every one, and the six subsections below move outward from the ingredient itself to the design choices around it. Section 11.1 isolates the target, the single ingredient the whole effect depends on, against random, learned and alternative hand-crafted alternatives. Section 11.2 varies where in the network the target is applied, how its weight is scheduled, and what form the classifier head takes. Section 11.3 asks whether one configuration survives changes of architecture and depth without retuning. Section 11.4 runs the classical alternative, the same bank placed in the forward path, and reports its failure mode. Section 11.5 tests whether a wider bank helps. Section 11.6 separates a second effect that the accuracy tables alone would conflate with the first: the prior also stabilizes optimization on backbones whose baselines collapse. Finally, Section 11.7 states what we did *not* vary.

11.1. The target is the ingredient

Table 20: a random fixed target of identical shape moves accuracy by at most 0.14 points, the mechanism is not "regression as regularization". A FitNets teacher trained on the same data at twice the cost does nothing anywhere we measured it (eight fractions, -0.25 to $+0.73$), making the free hand-crafted target strictly better than an expensive learned one in this regime. Among hand-crafted families, phase-invariant *magnitude* wins decisively; discarding magnitude scale (cosine loss) forfeits most of the gain, and heavily processed invariants are actively harmful targets.

11.2. Placement, schedule, and head

The tap depth is flat across the first three stages (within seed noise at both 1% and 10%) with a cliff only at the final stage: the one design knob in the study that is *not* a data-regime knob. The schedule start λ_0 *is* a regime knob (best 2.0 at 1–2%, 1.0 at 3–10%, 0.3 at 15–25%), but its reach is bounded: three attempts to push Δ past the measured feature gain by retuning λ_0 all failed, consistent with the law, the knob cannot manufacture G it can only realize it. Head-norm projection, derived from a traced scale degeneracy (the aux loss is invariant under feature-shrink with head-inflate, which collapses bottleneck ResNets), converts ResNet-50 from a bistable failure ($\{41.2, 36.4, 42.4\}$ across seeds) into a stable +3.9 ($\{44.5, 45.0, 44.4\}$) and is free elsewhere; we adopt it always-on. Notably, three plausible fixes tried *before* tracing the mechanism (gradient balancing, disabling automatic mixed precision (AMP), BatchNorm on the tap) all failed or backfired, the last making collapse $25\times$ faster, a small case study in mechanism-first debugging.

11.3. One configuration, every architecture and depth

A single champion setting transplants without retuning: at CIFAR-100 with 10% of the data, ResNet-18, -34 and -50 land +4.3, +4.0 and +3.9 against their own baselines (these are the head-norm cells of the backbone sweep, not the frozen-recipe envelope of Table 4); the new depth sweep confirms one λ_0 across ResNet-34 and -50 on six further domains (+0.4 to +9.0 at 5–10% everywhere except CUB's deep-scarcity floor, decaying on the right flank as the law requires). Head *form* is irrelevant where it plausibly might not have been: linear, cosine , and nearest-class-mean readouts measure the same aux-vs-baseline gap to within 0.3 points, the left-flank bottleneck is label information, not classifier expressivity.

11.4. The forward-path alternative

Placing the same bank *in the forward path*, concatenated fixed channels at the stem, the classical hybrid design, yields the study's sharpest structural contrast. Forward-path stems are the low-data accuracy leaders (energy-magnitude reaches +2.6 and +3.5 at CIFAR-100 1–2%, beating the auxiliary prior there), but every fixed forward variant we test, linear Gabor at two kernel scales, and four nonlinear energy families, pays a penalty band at 10–25% data (-1.4 to -7.3) that no filter choice escapes, and the phase-invariant variants never wash out even at 100% (-3.9): a hard constraint permanently occupies input bandwidth that abundant data would rather use. The auxiliary placement of the *same* knowledge is what removes the band, which is the design argument for fusing priors through training-only targets rather than through architecture.

11.5. Bank width

Widening the pinned 8-pair bank to 12 channels (via a third octave or two extra orientations, indistinguishable at 10×10 seeds) buys a real but local +0.3–0.5 at Tiny-ImageNet's 64 px and nothing at 32 px; across five further 64 px domains the pooled excess fails our pre-registered re-pin criterion. The pinned 8-pair bank therefore stays, with cross-domain evidence rather than convenience behind it.

11.6. Stabilization as a second, distinct effect

Across ConvNeXt (under the frozen SGD recipe), ResNet-50 (without head-norm), Swin-T (on five datasets), and Swin at ImageNet64, baselines are bistable, some seeds train, some sit at chance, while the prior arm trains tightly every time. Swin's feature-level variance shows the instability lives in the *features* (baseline evaluation σ up to 4.3 vs. 0.2 with the prior). We report stabilization as a real, separate property; we do not claim routine variance reduction (a 10×10-seed test of that claim came back negative, and we report that too).

11.7. What we did not ablate

Completeness cuts both ways, so we list the design constants that the method declares and this paper does *not* vary, and separate those that could change a conclusion from those that cannot.

Four are held fixed by construction and are applied identically to both arms of every pair, so they bound absolute values without being able to explain a paired difference. The training recipe itself is the instrument and is never tuned (Section 3.1). The target is pooled to the tapped stage's resolution by average pooling; max or learned pooling are untested. The auxiliary head is a single 1×1 convolution, and although we ablate the *classifier* head thoroughly (Section 11.2), we never widen or deepen the auxiliary one. The linear evaluation uses a fixed ridge constant of 10^{-4}; it shifts every arm's absolute accuracy together, so G's sign and ordering are unaffected, but the numeric value of G is conditional on it.

Three are more consequential, because they define the knowledge source rather than the measurement. The bank's envelope width is tied to its frequency by a fixed relation, so every filter carries the same number of cycles; we vary how many scales and orientations the bank holds (Section 11.5) but never that constant, so "how localized should an oriented-energy target be" is open. The target is computed on BT.601 luminance, which makes the prior colorblind by construction, and a color-opponent target is untested. And while the endpoint of the weight schedule is ablated and load-bearing ($\lambda(T) = 0$ against $\lambda(T) = 0.1$), its cosine *shape* is not; a linear or stepped decay to the same endpoint may behave identically, and we do not know.

None of these is a confound for the results reported here, since a paired Δ and a same-protocol G gap are differences taken under identical settings. They are the boundary of what one bank, one head form and one schedule shape can establish, and the first two are the natural next dimensions for anyone extending the taxonomy to other structural priors.

Table 21
What to use when: the grid distilled to recommendations. Costs are training-compute multiples of the baseline; gains are typical measured ranges, not promises. "Prior" is the champion MomentAux configuration verbatim.

Regime	Recommendation	Cost	Measured basis
Conv backbone, photo-like data, mid band (2–15%), 2× affordable	SimCLR init	2×	+2..+9 over baseline; beats prior by +2..+5
Conv, photo-like, 2× *not* affordable	prior	1.02×	+1.4..+6.7; ∼SSL parity at extremes
Conv, extreme scarcity (≤1–2%)	prior (or fwd stem)	1.02×	SSL margin ≤1 pt; forward stem best ≤2%
Satellite-like statistics, low data	prior	1.02×	beats SimCLR at 7 of 11 EuroSAT fractions
Any ViT trained the modern way (DeiT aug)	prior *and* augmentation	1.02×	flip: beats SSL at every fraction at 2×; at 5× only in the higher-data band (both populations); +14..+25 over recipe alone
ViT and Swin from scratch, any scale	prior	1.02×	+3.2 at 1.28M imgs; +13 and +26 at 224 px; stabilizes Swin
ImageNet-transferable task	transfer; **never** add prior	–	tax up to −17, feature-side
Domain-shifted from ImageNet (e.g. stains)	prior from scratch competitive	1.02×	transfer advantage shrinks to +2..+3
Data sufficient (≥50%)	nothing (all converge)	–	every intervention within ±0.8

12. A decision guide

Table 21 is the benchmark's practical distillate. Its strongest single row is the modern-ViT one: for anyone training a small or mid-sized ViT with the standard recipe, a fixed spectral auxiliary target at 2% overhead is, in every regime we measured, either the best or tied-best intervention available, and it composes with the recipe rather than competing with it.

13. Discussion

What the decomposition buys, and what it does not. The grid's size is not its point; its point is that a single low-dimensional structure organizes a large part of it. We are careful with "explains": binned on baseline accuracy the readout term has $R^2 = 0.18$ and a residual spread of 2.2 points, so this is an account of a trend and of a *sign*, not a predictor of a cell's value. The decomposition in Eq. 1 says that an intervention's visible benefit is feature gain passed through a readout channel whose behavior is set by how well the task is already being solved. This explains, with one mechanism, observations that otherwise require separate stories: why gains collapse at extreme scarcity even when features improve (readout deeply negative, five labels per class cannot express a better representation); why relabeling the same pixels with coarser classes triples the visible gain (the baseline crosses the readout zero, not because features changed; we measure them unchanged); why every intervention converges at sufficiency (both terms go to zero); and why coarse label spaces read $\sim$2 points more gain under coarse evaluation than fine on the same checkpoints (the measuring stick itself is label-hungry).

What the currency account contributes to fusion theory. We want to be careful about what is new here, because none of our three outcomes is. Fusion has distinguished complementary from redundant sources since Durrant-Whyte [19], and stack and substitute are that distinction; the destructive third case is catastrophic fusion [23], negative transfer [24] and modality competition [25] under three different names. What the classical taxonomy does not provide, because it was written for sensors whose relationship is known from the sensing geometry, is a way to *determine* which case holds when the sources are a hand-crafted prior and a learned representation, whose relationship is not evident a priori and, as Section 6 shows, is not predictable from their family names either: the same prior is redundant with effective self-supervision and complementary with augmentation. Our contribution is therefore operational rather than taxonomic. Fusion adds value exactly when the sources' contributions are non-overlapping *in feature space*, that overlap is measurable in advance by a cheap linear evaluation of each source alone, and the measurement predicts the combination. Fusion decisions can be made by measurement rather than by fashion, before any joint training is run.

On the third outcome our addition is a magnitude law rather than a phenomenon. The literatures above establish that combining sources can hurt; what we measure is that the damage is *proportional to what the displaced source had supplied*, is carried by the feature term rather than the readout, and falls to a measured zero exactly where the partner had supplied nothing. That makes it a fusion outcome on the same currency logic as the other two rather than a separate pathology, and it is the part of the taxonomy we would defend as ours.

What the attention result does and does not say. The persistent, scale-growing ViT deficit is a statement about *training small-data vision transformers from scratch*, where we show a fixed spectral target supplies structure that neither 1.28M images nor the DeiT recipe nor 2×-compute SSL supplies as cheaply. It is not a claim about web-scale pre-training, where data eventually buys everything (our own convolutional cells show exactly that limit). The right reading is deployment-oriented: wherever from-scratch ViTs are trained on 10^4–10^6 images, medical imaging, remote sensing, industrial inspection, a 2%-overhead structural prior is close to free accuracy, and the smaller-model-plus-prior configuration can beat the larger model outright.

On being wrong in public. Four of our major pre-registered predictions missed: SSL did not collapse at 1% on convolutional backbones (it won everywhere under plain augmentation); augmentation did not substitute for the prior (it amplified it); stronger contrastive views did not help SimCLR (they hurt it); and an early "gain tracks deficit" law was falsified outright and retracted. We report these because the misses carry as much information as the hits, each redirected the program, and because a benchmark that cannot be wrong is not measuring anything.

14. Limitations and future directions

The law's status and domain of validity. The law is an *empirical* regularity, not a theorem, and we state its measured scope plainly so that it is not read as more than it is. It was fitted to nothing: we report the measured readout curve and use it directly. Its demonstrated domain is 13 datasets across six visual domains, backbones from 5.7M to 86M parameters, images from 32 to 224 px, and training sets from 500 to 1.28M images, all under one supervised recipe with an auxiliary prior trained from scratch. Cells initialized from self-supervised or ImageNet weights lie outside the scope the law was derived under and are scored separately. Within that domain its credibility rests on prediction rather than fit: bands registered before the runs called an unseen backbone family's feature gain from baseline heights alone (four of four in band), eight cells on four unseen domains (seven in band, eight of eight correct in sign), and the ImageNet-scale residual. Where the law is weakest is equally explicit: on the right flank it predicts readout ≈0 and gets it, which is true but not discriminating, so 487 of 1,100 in-scope cells cannot test it at all, and of the 511 that can, those above the crossing are right only 61% of the time. Nothing here licenses extrapolation to web-scale pre-training or to architectures whose baselines do not train under this recipe. The one non-classification test we run, semantic segmentation (Section 10), confirms the law's positive branch on a different task and a different metric and cannot test its negative branch at all, for the reason given there.

Known boundaries. Two are worth naming. The first is regime: the sign prediction is right 88% of the time below the crossing and 61% above it, so the account is informative where the readout term is large and close to uninformative where it has decayed to nothing. It should be read as an account of the scarce-data flank. The second is a recurring exception concentrated on Food-101 and PathMNIST at $\geq$20% data, where the prior still improves frozen-feature quality while costing end-to-end accuracy. The account has no term for this "better features, worse accuracy" regime; modeling it, plausibly an interaction between late-stage optimization and sufficiency, is the clearest open problem the grid poses.

Scope of the instrument. The frozen recipe is, in effect, a ResNet recipe: architectures that cannot train under it (ConvNeXt, Swin under SGD) enter only through diagnostic AdamW cells, and their bistability, itself a finding, limits headline claims. PathMNIST's shifted test split confounds its high-data cells for every method. Linear evaluations lose interpretability at 100% cells (the evaluation ceiling) and at ImageNet64 are budget-bound by construction. Both ImageNet stages were subsequently run across 1–25% as well as at full data, so the envelope and not only the right flank is measured at scale; a 1% ImageNet64 cell is 12,811 images over 1000 classes, thirteen per class in a label space five times finer than anything else here. That is the scarce, fine-grained corner where the readout term is largest, and it cost us a registered prediction: we called an interior peak at 5–20% and the convolutional envelope is instead monotone falling from the smallest runnable fraction, because 1% of ImageNet64 is already 12.8k images — past the peak of every smaller population. The envelope is not absent at scale; its peak sits below anything this dataset can express. One prior family (Gabor and moment energy) is tested as the knowledge source; the taxonomy predicts, but does not yet demonstrate, that other structural priors will slot into the same currency logic.

Future directions. Three follow directly. (i) *The exception cluster*: instrument the late-training dynamics of high-data Food-101 and PathMNIST cells to locate where good features fail to become accuracy. (ii) *Other currencies*: repeat the fusion matrix with priors carrying genuinely different structure (depth, symmetry, frequency-phase) to test whether currency-overlap measured by linear evaluation predicts stacking in general. (iii) *Attention at deployment scale*: the model-scale trend (+13 rising to +26 from ViT-S to ViT-B at fixed data) invites the obvious next point on the curve, and the stabilization-plus-structure account suggests trying the prior as a warmup for large-ViT pre-training proper.

15. Conclusion

We opened with one intervention worth +26 points, nothing, and −17 points in three different settings, and asked what distinguishes the cases. The answer this study supports is that information sources trade in currencies, that what a source is worth is set by which currency is scarce rather than by how modern the source is, and, the part that makes it useful rather than merely descriptive, that each source's currency is measurable on frozen features from that source *alone*, before any combination is trained. Same currency, and the second source is redundant however sophisticated it is; different currencies, and the gains compound; a shaping prior poured into an initialization that already carries mature features destroys them in proportion to what they were worth.

We reached this through an instrument rather than a set of experiments: one recipe, fixed data, pinned filters, declared comparator budgets and pre-registered predictions, across 13 datasets, 9 backbones and four orders of magnitude in data. The measurements organize under one empirical decomposition separating feature gain from readout, which predicted an unseen backbone family's feature gain in advance and tied accuracy to features within about a point at ImageNet scale, and whose sign holds on 79% of testable cells overall and 88% in the scarce-data regime it describes best. The same discipline cost us two accuracy claims, which the comparator-budget experiment overturned and which we restate here as cost claims.

The most surprising thing the instrument found is the direction of the attention result. A bank of oriented filters that predates deep learning is worth +13 points to ViT-S/16 and +26 to ViT-B/16 at 224 px, it is worth *more* to the larger model, and it is still worth +3.2 after 1.28 million images, in a regime where our convolutional baseline is neutral to within 0.04. Whatever small transformers fail to learn about oriented structure at these scales, more parameters and more data did not supply it, and 2% of a training run did.

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CRediT authorship contribution statement

Ahmad AlMughrabi: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing, original draft, Writing, review & editing, Visualization, Project administration. **Albert Clop**: Methodology, Supervision, Writing, review & editing. **Benjamin Busam**: Supervision, Writing, review & editing. **Ricardo Marques**: Supervision, Conceptualization, Writing, review & editing. **Petia Radeva**: Supervision, Conceptualization, Resources, Project administration, Funding acquisition, Writing, review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All artifacts needed to reproduce every number in this paper are released at `https://github.com/GCVCG/MomentAux`: the training and evaluation harness (configs for all 2,978 cells, the subset indices, the fingerprinted filter banks, the single training command, the aggregation and audit scripts), and the complete result tables in open formats. The tagged release additionally carries the *raw* evidence as downloadable assets: every run's `final.json` record (configuration as executed, accuracy, parameter and floating-point-operation accounting, environment), every linear-probe record, every per-epoch training curve, and the campaign logs, with `SHA256` checksums. The latter comprise a per-cell table (`all_results.csv`: 2,978 rows with dataset, backbone, intervention, fraction, seed count, accuracy mean and standard deviation, probe accuracy, paired baseline, Δ, its standard error, G and readout), a configuration-by-fraction view (`results_by_portion.csv`), and a workbook combining both with a column dictionary and the law audit. Every table and figure in the paper regenerates from these files by one command, and the sign-law audit of Table 7 is itself a released script rather than a manual query. Public datasets are cited in Section 4.1; the subset indices we release make the exact image selection reproducible without redistributing the images. Model checkpoints (223 GB) are not distributed; every cell retrains from its released configuration.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) in order to improve the language and readability of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Table 22
SimCLR initialization ($2\times$ compute) minus the prior ($1.02\times$) on CIFAR-100 with ResNet-18 under plain augmentation. The margin is unimodal: SSL's advantage is a mid-band phenomenon that vanishes at both extremes. A positive entry is a fraction where the $2\times$-compute initialization wins.

Frac.	SSL − prior	Frac.	SSL − prior
SimCLR at $2\times$ compute; at $5\times$ the margin is $+3.1$ to $+9.2$ throughout			
1%	+0.85	10%	+5.01
2%	+2.38	15%	+4.09
3%	+2.32	50%	+0.58
5%	+3.90	100%	+0.05
7%	+4.99		

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Appendix

The appendices hold the material a reader may want to check but which would interrupt the argument. Appendix A gives the self-supervised margin at every measured fraction; Appendix B the component curves (G and readout) that the law decomposes; Appendix C the controls cited in passing in the main text; Appendix D the per-domain notes, including the PathMNIST confound; Appendix E the dataset table; Appendix F the reproduction commands and released artifacts; and Appendix G illustrative panels for the feature effect measured in Section 5.5.

A. The self-supervised margin

The prior's own envelopes are in the main text (Tables 4, 5 and 6). Table 22 gives the remaining envelope those tables summarize in prose: the margin of a $2\times$-compute SimCLR initialization over the prior at every measured fraction. Every entry is a paired difference against a shared baseline arm under the frozen recipe.

B. The law's components

Feature-gain curves. G is non-monotone in data and dataset-specific in shape: CIFAR-10 4.81, 5.52, 3.97, 0.65 at 0.5, 1, 2.5, 5k images; CIFAR-100 3.88, 5.33, 6.35, 3.66; Tiny-ImageNet falls monotonically 4.33, 2.67, 1.70 (1, 5, 10k); STL-10 4.70, 4.98, 3.20. Wherever the readout term is flat, the end-to-end envelope peak coincides with the G peak; the envelope's shape *is* the feature-gain curve.

The readout branch. Mapped on ceiling-free CIFAR-10: +1.80 at baseline 39.4, +1.14 at 51.7, +0.44 at 69.1 and at 80.7: positive throughout and decaying, never growing, toward sufficiency. Below the crossing, measured values reach -2.8 (base 8.9) and -2.7 (base 5.3). The crossing bracket $[31.8, 40.3]$ is pinned by a Swin cell still negative at 31.8 and a conv cell already positive at 40.3.

Label-space crossings. On byte-identical CIFAR-100 pixels, a full 2×2 of training label space $\times$ evaluation label space (100-way vs. 20-way) moves G by at most 1σ: invariance. On Tiny-ImageNet the same crossing finds two real, opposing effects (evaluation-space +1.9, checkpoint-set +1.6, both $\sim5\sigma$) that cancel on the diagonal; G comparisons in the paper therefore always fix the evaluation space.

C. Selected controls

Contamination. Re-evaluating the low-data CIFAR-100 cells on ciFAIR-100 [36], which replaces the test set's 927 near-duplicates, moves the prior's gain by at most 0.20 points at any fraction; both arms eat identical contamination, so it cancels in Δ.

Granularity, decisively. Relabeling CIFAR-100's fixed subsets with their 20 coarse labels (byte-identical pixels, identical steps) reproduces the full gain at matched data (+5.84 vs. +5.30 at 5%): gain follows data and task performance, not class count. The mirrored Tiny-ImageNet control (positional, visually incoherent coarse groups) is the cell that falsified our own earlier account: class count dropped tenfold, baseline accuracy did not rise, and no readout boost appeared: exactly as Eq. 1 requires and as "label count drives gain" does not.

Robustness. On CIFAR-100-C [50], corruption robustness tracks 55–60% of the clean gain at low data and is neutral at 100%; the prior neither buys nor costs robustness beyond its accuracy effect.

D. Domain notes

EuroSAT: the prior beats SimCLR at 7 of 11 fractions (materially +0.8–+1.9 in the 1–10% band): consistent with SimCLR's photographic view-invariances transferring poorly to satellite statistics while the spectral target is domain-agnostic. DTD (texture) is the reverse: SimCLR dominates from a few hundred images. Food-101 is SSL territory on conv backbones. PathMNIST declines with added data for *every* method above ∼10–15% (baseline 92.0 to 86.7 from 10% to 100%), the signature of its center-shifted test split; we therefore interpret only its low-data cells. The domain axis thus does not split "photo vs. non-photo"; each domain's statistics decide which currency is scarce.

E. Datasets

Table 3 in the main text lists every dataset with its training split size, class count and resolution as trained, and we do not repeat it here. Five visual domains are represented; the two ImageNet stages supply the data-scale and the resolution-and-model-scale axes respectively. Sources with variable native resolution are squash-resized once, offline, and the resized copies are what every arm consumes. The grid also contains five relabeled controls derived from CIFAR-100 and Tiny-ImageNet, which share their parents' pixels exactly and differ only in the label space; they are listed at the foot of that table and are what make the dataset-identity count in Table 7 exceed the number of image sources.

F. Reproducibility

One entry point trains any cell (`train.py -config <cell> -seed N`); one command regenerates every table from run records; one command re-runs the sign-law audit. Subset indices, filter-bank fingerprints, and per-run environment records (library versions, worker counts) are released with the runs they describe. Training refuses to overwrite a completed cell, holds an exclusive per-run lock (two of this study's incident classes made these guards necessary), and writes checkpoints atomically; mirrors are verified by *loading and evaluating* checkpoints against their recorded accuracies, not by file listing. The full harness, configs, subsets, banks, runner, aggregation, audit, is released with the paper at the repository cited on the title page; an archival DOI will be minted on acceptance.

G. Illustrative panels

The measurements behind the prior's feature effect are the alignment statistic and the linear evaluation reported in Section 5.5; the two panels here illustrate them. Fig. 9 is *selected* on test images the prior arm classifies correctly and the baseline does not, so it shows what the difference looks like where it is present, not how often it is present. Fig. 10 is not selected, and is included precisely because it shows the effect less clearly than the numbers do.

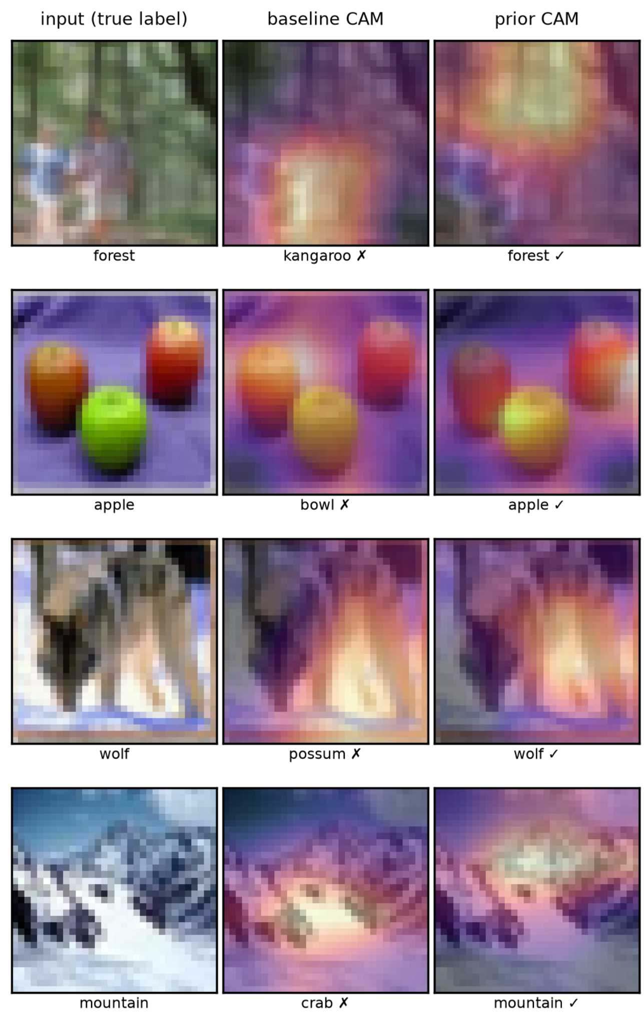

Figure 9: Class-activation maps, CIFAR-100 at 5%, for four selected test images (prior arm correct, baseline wrong). The prior arm's evidence concentrates on object contours and texture boundaries where the baseline's is diffuse at this data scale.

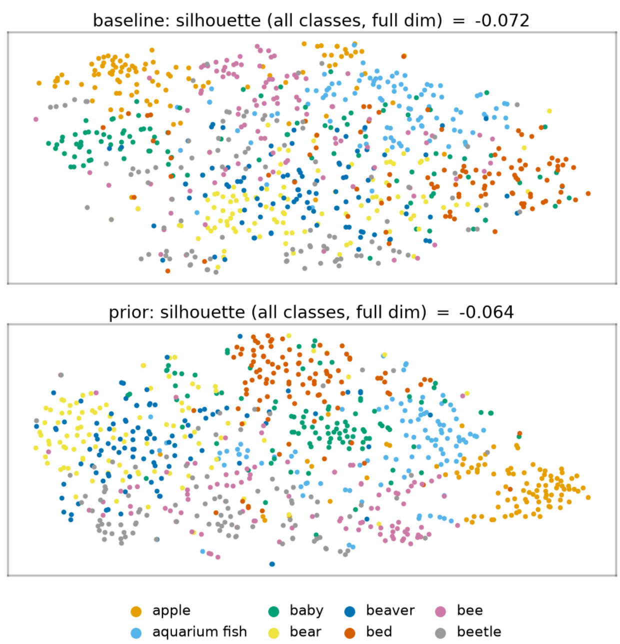

Figure 10: t-SNE of frozen penultimate test features, first eight classes, baseline (top) and prior arm (bottom) at the CIFAR-100 5% cell. The silhouette in each title is computed over *all* classes in the *full* feature space, not on the two projected dimensions shown. Reported with its own limitation: the silhouette score over all classes at full dimension moves only from -0.072 to -0.064, so this projection does *not* show the $+6.3$-point feature gap that the linear evaluation measures at this cell. We include it because a reader would otherwise reasonably wonder whether it does.